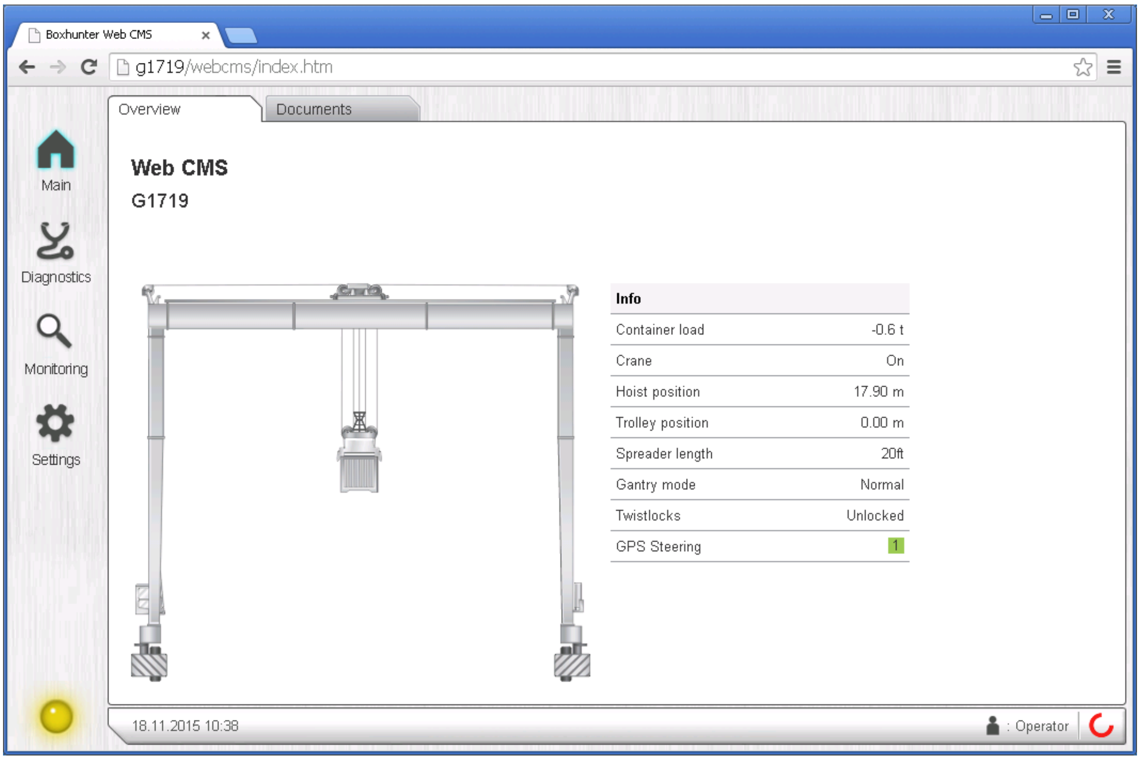
OWNER'S MANUAL

WEB CMS



ORIGINAL INSTRUCTIONS
XDMAP948529en / A / 20 May 2016

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1 INTRODUCTION

1.1 Overview

This document serves as the owner's manual and a quick reference guideline for the installation and configuration of the Web CMS system. The document presumes that all electrical connections have already been completed when the Web CMS installation begins. The installation and configuration is separated into the following steps:

1. *Connecting the Beckhoff C6915 system.*

   This step is described in *Connecting the Beckhoff Industrial PC (page 26)*.
2. *Flashing the Web CMS software onto the CF memory card used in the Beckhoff C6915 system.*

   This step is described in *Flashing the CF card with the Web CMS software (page 28)*.
3. *Connecting the PC to the crane network.*

   This step is described in *Connecting the PC to the crane network (page 31)*.
4. *Testing Web CMS with the PC's web browser.*

   This step is described in *Opening Web CMS on the PC (page 35)*.

1.2 Copyright notice

1.3 Contact information

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During the warranty period, please send questions and claims regarding the product warranty to ports.warranty@konecranes.com.

2 PRODUCT DESCRIPTION

2.1 Basic description

Web CMS is a web application that displays diagnostic, operation, and usage data from the crane. The software reads data from the PLC. The Web CMS hardware is located in the electrical cubicle of the crane. The web application can be accessed locally directly from the crane or from a remote location via a wireless connection.

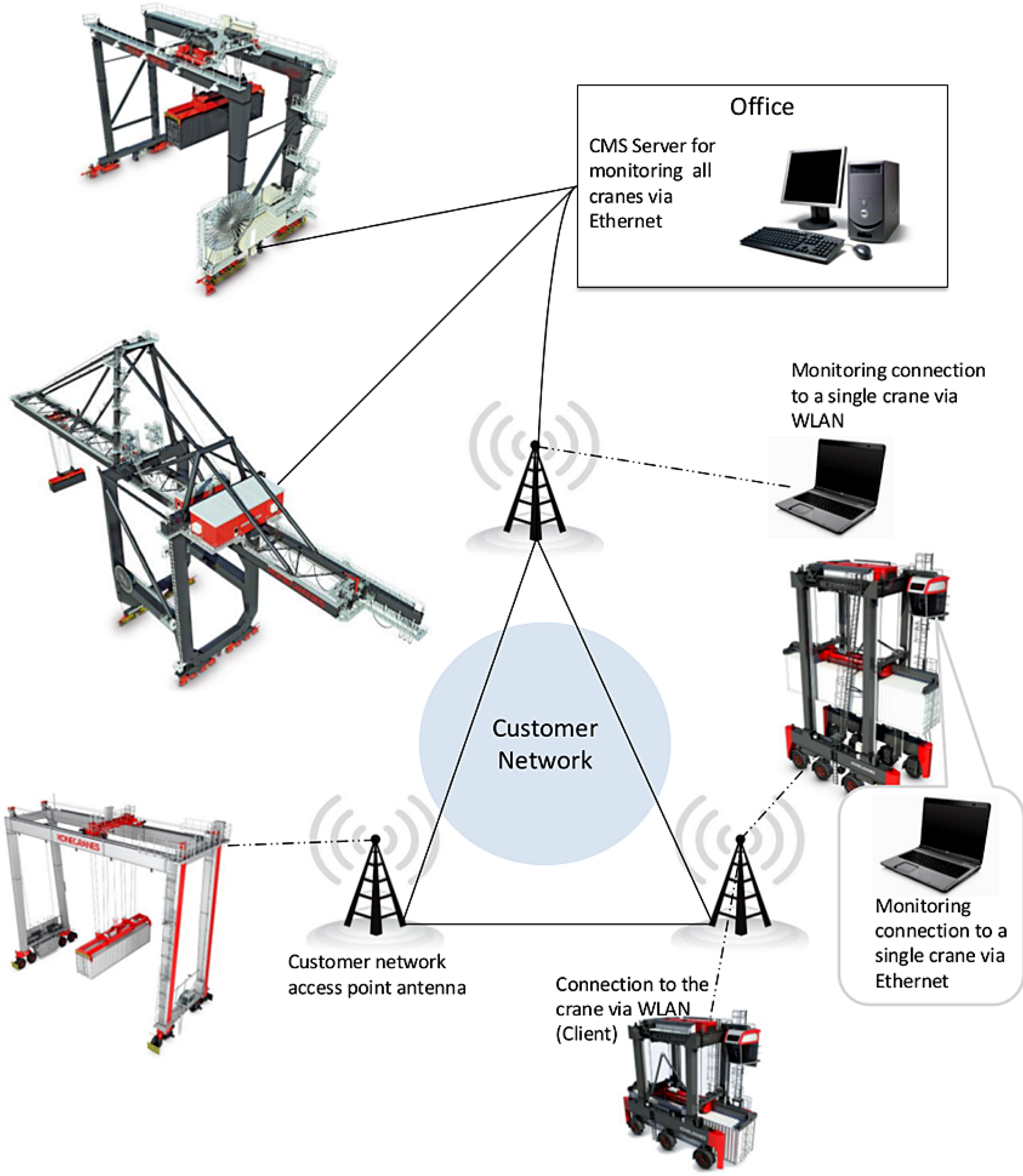


Figure 1. Web CMS system

CMS Server	CMS Server is the central system for all the Web CMS instances. From CMS Server, it is possible to view the status of each individual crane that has CMS or Web CMS running.
Web Service	The Web CMS application communicates with the PLC using Web Service. Web Service runs on the same hardware as the Web CMS application.
PLC Interface	Web Service reads the Web CMS data from a specific memory area or data block within the PLC. This block is usually DB56.
Data Logger	The Web CMS device also contains Data Logger which gathers data from the crane. The data is sent to CMS Server when the connection is available. Otherwise the data is buffered to the device and sent later on. The logged data can be viewed from the Reporting section of CMS Server.

2.2 System configuration

Software	The Web CMS application is a web application that can be viewed using a common web browser. The dynamic html pages are served by a web server located in the crane. The application uses AJAX to update the html page values asynchronously.
Network	The Web CMS hardware provides access to the Web CMS pages from two separate networks. The first network is the crane's private network. The HMI and the PLC are on the same network. Access to this network requires direct Ethernet connection to the crane. The second network is a public one and can be accessed via a wireless connection. Remote connection to the crane is limited to the Web CMS pages only.

2.3 Technical requirements

The following table lists some technical requirements that must be considered when using Web CMS.

Connection	A continuous connection to the crane is required to use Web CMS. The html pages are updated sequentially, and if there is a communication failure, the pages cannot be accessed.
Client software	Web CMS can be used with a common web browser. Web CMS can be used with most web browsers, but officially only the two latest versions of Internet Explorer, Google Chrome and Firefox are supported. The minimum recommended resolution for Web CMS is 1024 x 768 pixels, but the software also supports other resolutions. The application uses JavaScript to build and update the html pages. It is important that JavaScript is enabled in the web browser. Web CMS does not work without JavaScript.

3 WEB CMS PAGES

The Web CMS application consists of four pages:

- Main page
- Diagnostics page
- Monitoring page
- Settings page

Access the Web CMS pages by entering the IP address of the crane in your web browser. You must have a connection to the crane to use Web CMS.

3.1 Main page

The *Overview* tab on the *Main* page shows the most relevant online information about the crane. This information consists of the crane identification number and active status values (for example, load, position and spreader length). The crane state and the date and time on the PLC are visible in the page footer on all the Web CMS pages.

The crane state indicator colors are:

- Green – No alarms or faults active on the crane
- Yellow – At least one alarm is active, no faults
- Red – At least one fault is active on the crane
- Gray – Crane state is unavailable (for example, PLC is unavailable)

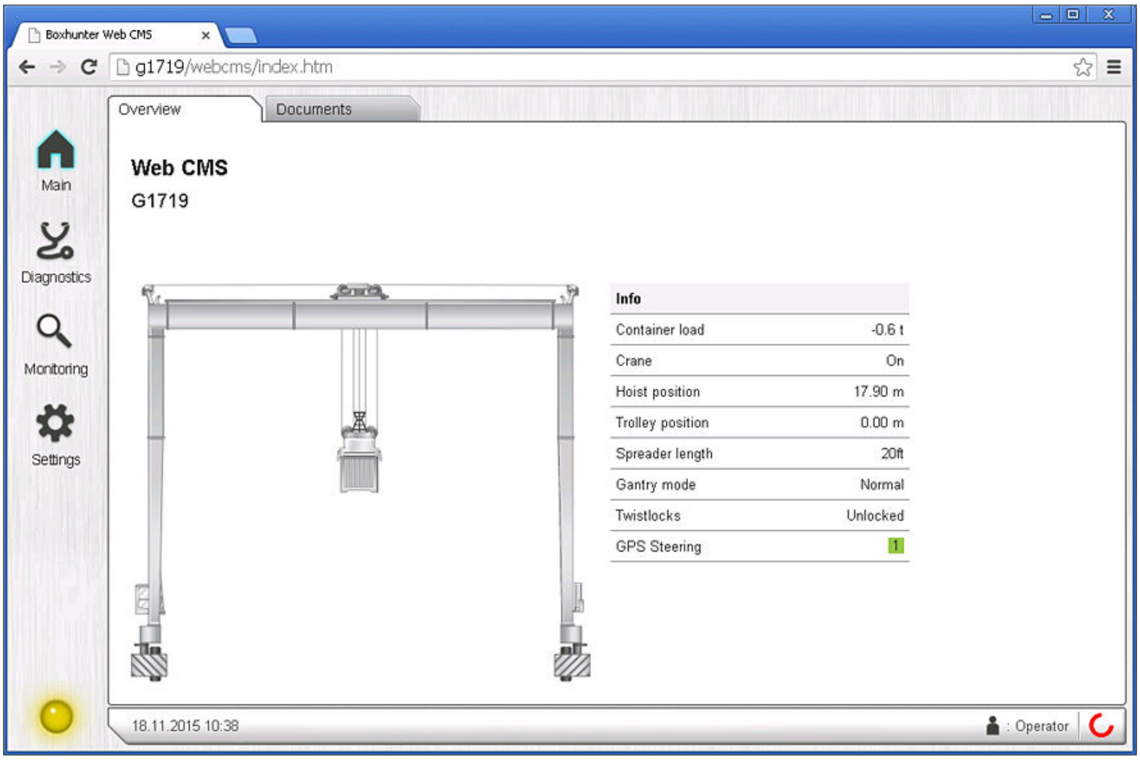


Figure 2. Overview tab

The *Documents* tab shows the document files that are uploaded in the Web CMS system. Typical documents include technical drawings and operator's manuals.

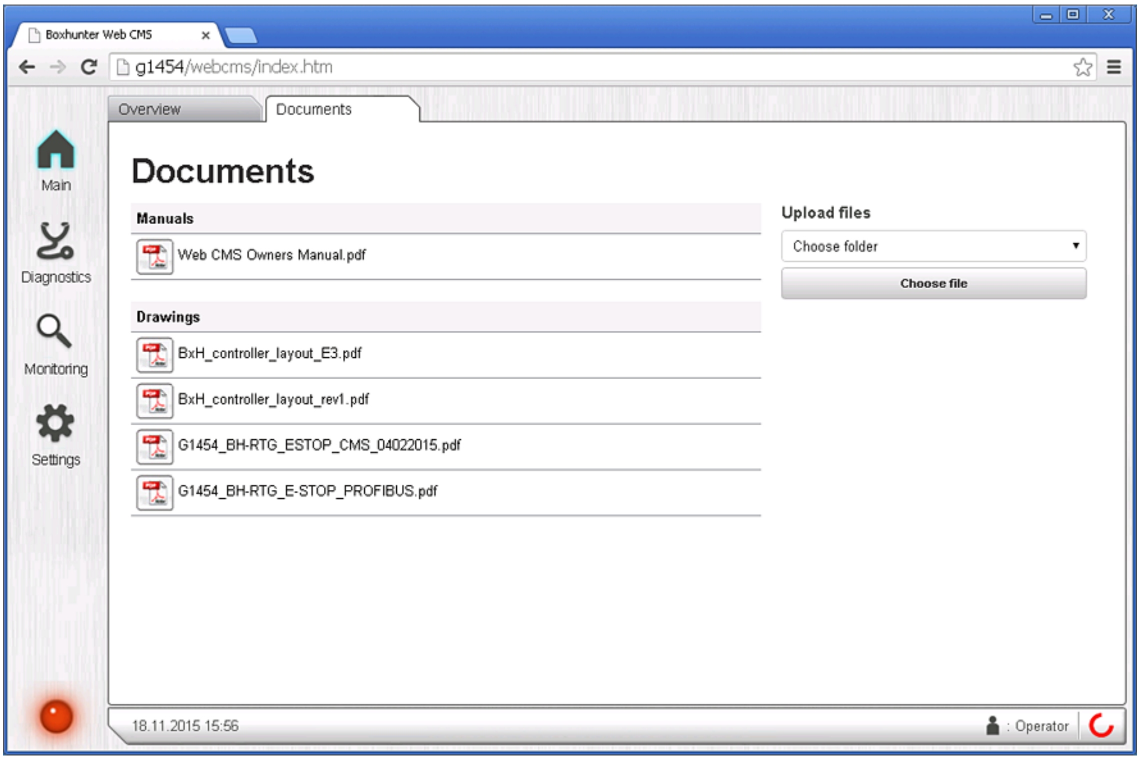


Figure 3. Documents tab

3.2 Diagnostics page

Active messages and message history can be found from the *Diagnostics* page of Web CMS. You can filter the message list according to message class by pressing the **Alarms**, **Faults**, **Events**, and **Bypass** buttons. To get additional information about a message, press the message row. The *Message history* tab has also extra search and filter options available, such as text search and searching by a date range.

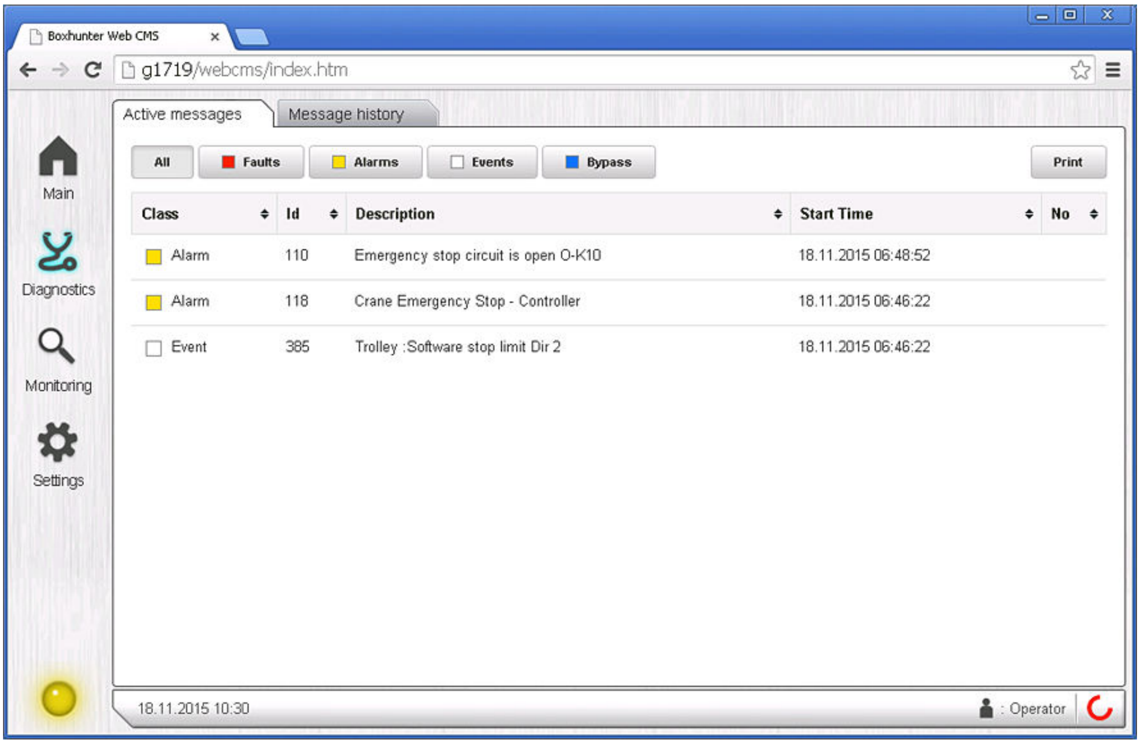


Figure 4. Active messages tab

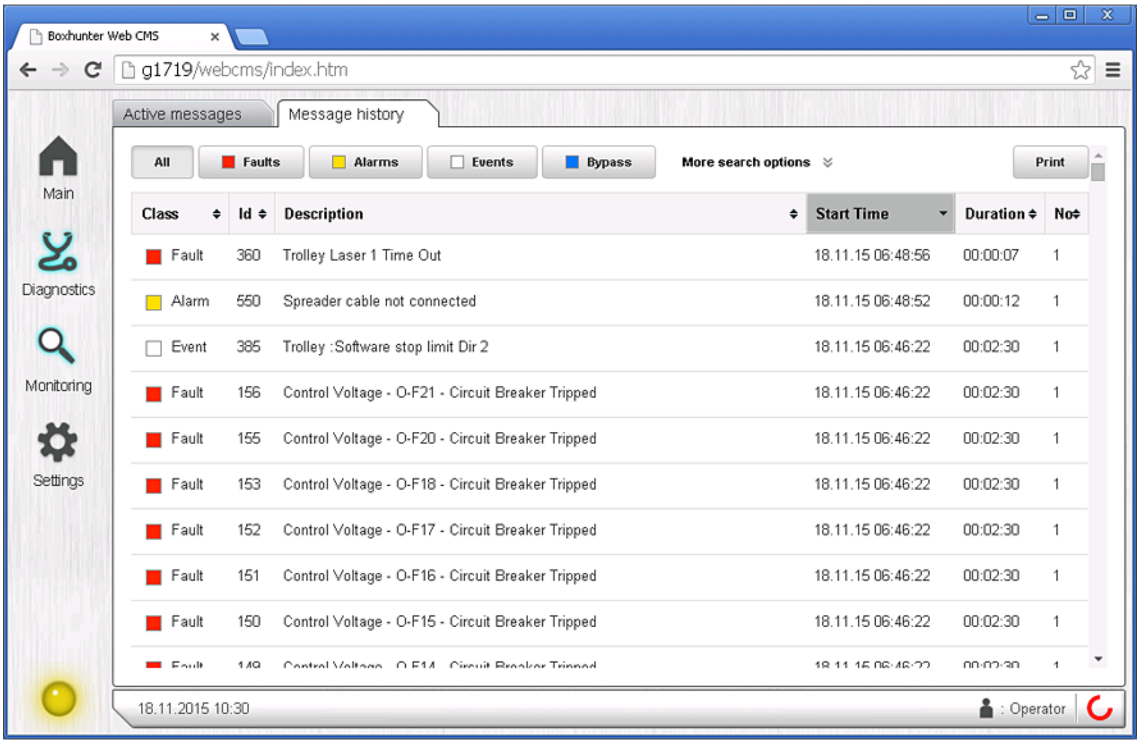


Figure 5. Message history tab

3.3 Monitoring page

The *Monitoring* page provides detailed information of the crane.

3.3.1 Hoist tab

The *Hoist* tab of the *Monitoring* page provides detailed information, such as limit states, position values, speed signals, load measurements, and condition monitoring values of the hoisting machinery.

Hoist Limits

The hoisting machinery limit states and position values can be seen on the *Hoist Limits* tab of the *Monitoring* page.

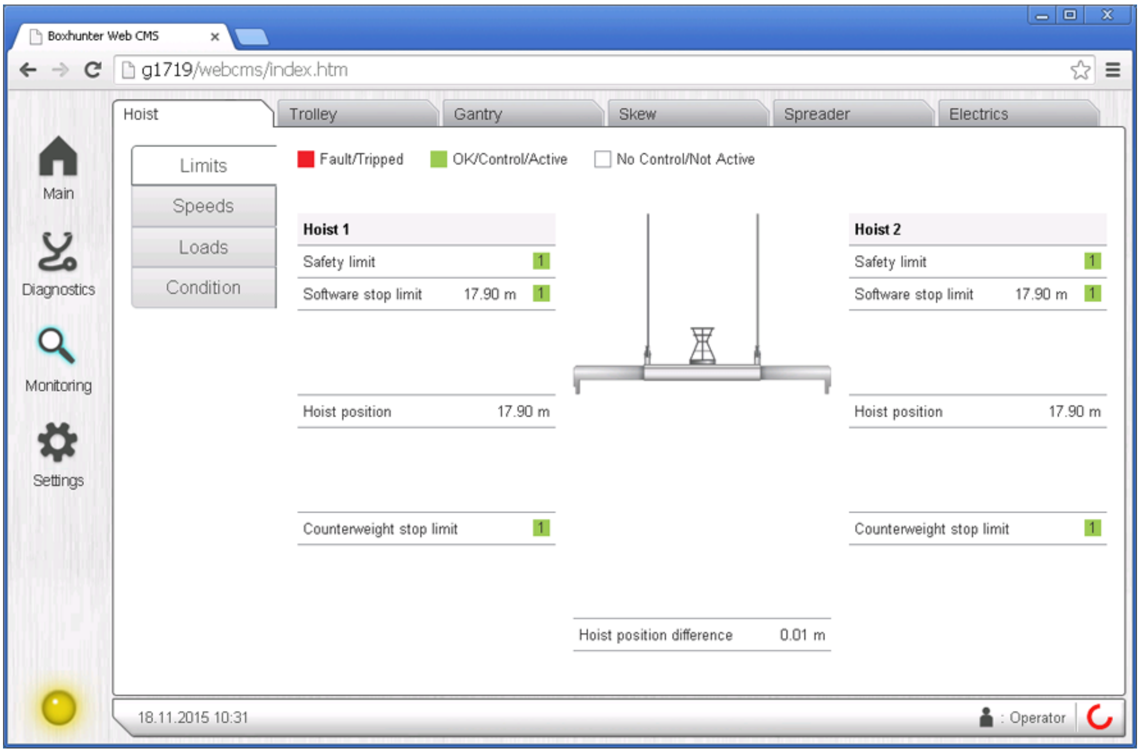


Figure 6. Hoist Limits tab

Hoist Speeds

The hoisting machinery speed signals are shown on the *Hoist Speeds* tab. The information can be used to diagnose unexpected slowdowns, or situations when the machinery is prevented from running at all (or runs in only one direction).

The speed signal proceeds from left to right, from the controller to the PLC, and then on to the drives and motors. Active direction bits, disable bits, brake contactor state, and fault bits are also displayed for diagnostics purposes.

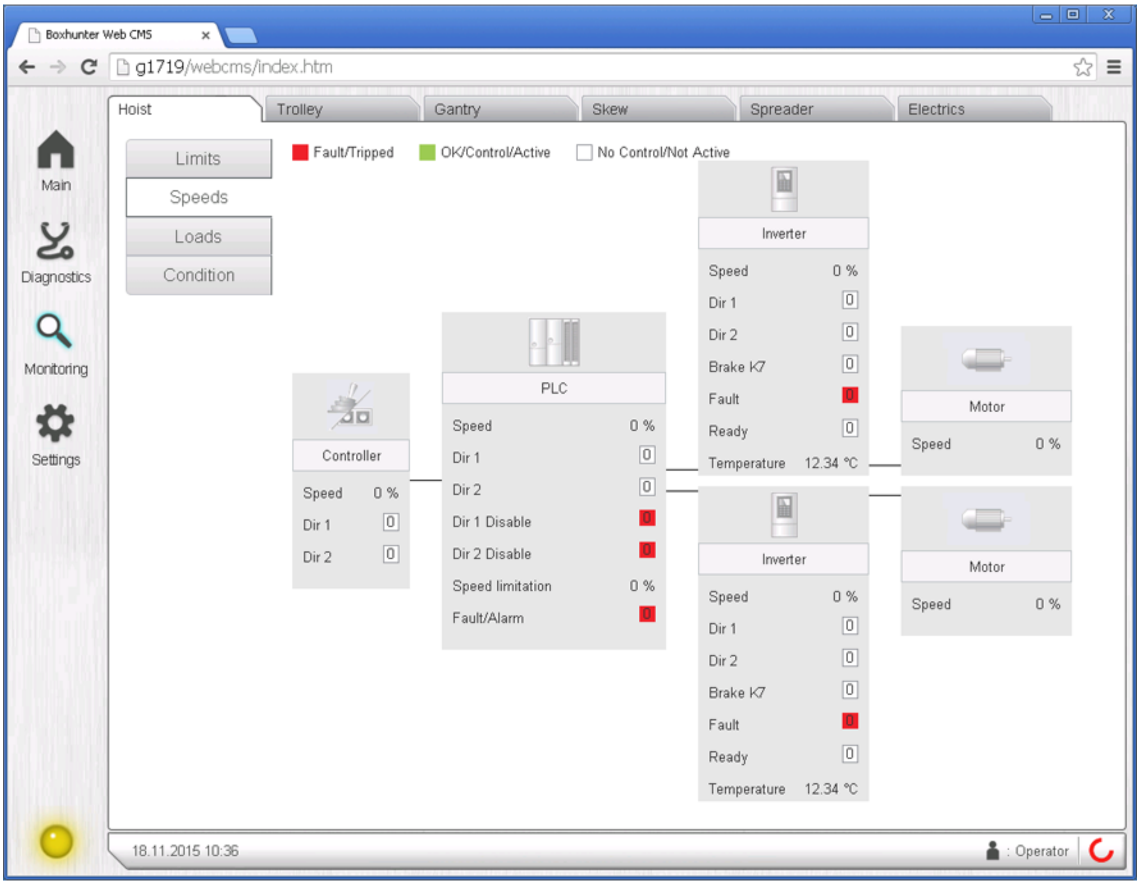


Figure 7. Hoist Speeds tab

Hoist Loads

Hoist load measurements are shown in the *Hoist Loads* tab of the *Monitoring* page.

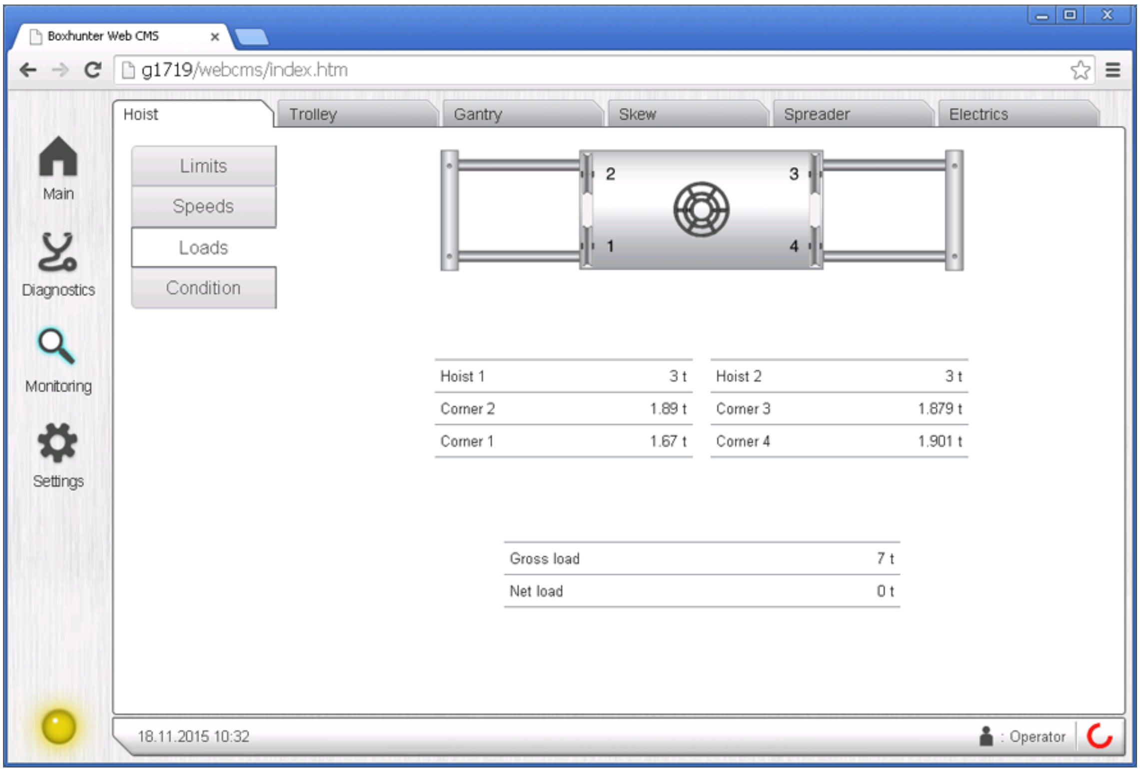


Figure 8. Hoist Loads tab

3.3.2 Condition monitoring

Condition monitoring values are available for hoist, trolley, and gantry machineries in the *Hoist Condition*, *Trolley Condition*, and *Gantry Condition* tabs of the *Monitoring* page.

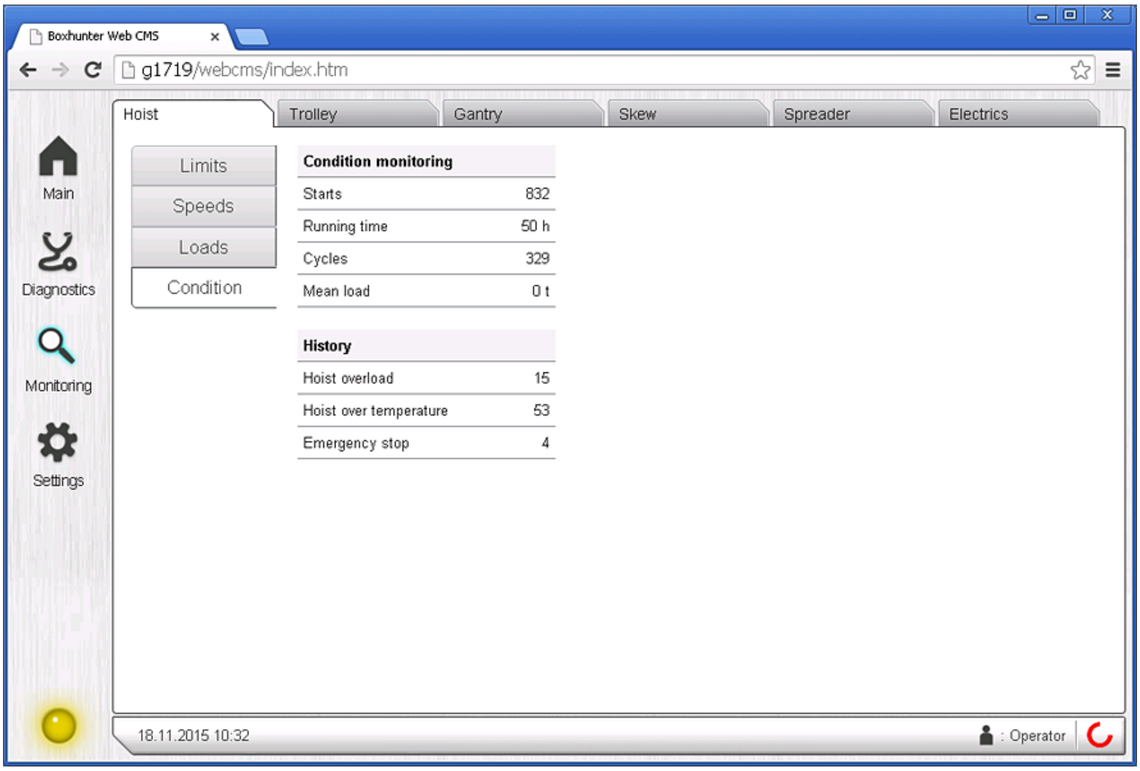


Figure 9. Hoist Condition tab

3.3.3 Trolley tab

Trolley hardware and software limit states and trolley position values are shown on the *Trolley Limits* tab of the *Monitoring* page.

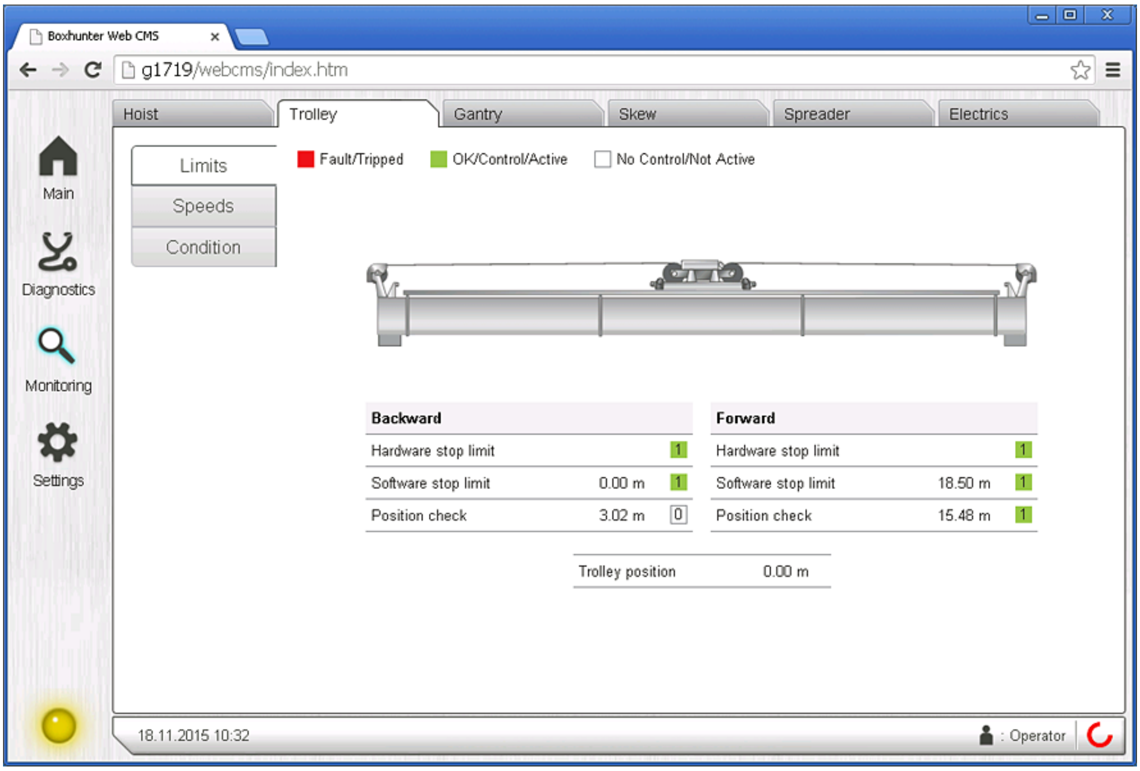


Figure 10. Trolley Limits tab

The *Trolley Speeds* tab works in the same way as the *Hoist Speeds* tab, but with only one drive and one table for the motors.

3.3.4 Gantry tab

The *Gantry* tab of the *Monitoring* page provides information of the gantry machinery.

Gantry Limits

The *Gantry Limits* tab displays the states of various slowdown and stop limits related to the gantry movements.

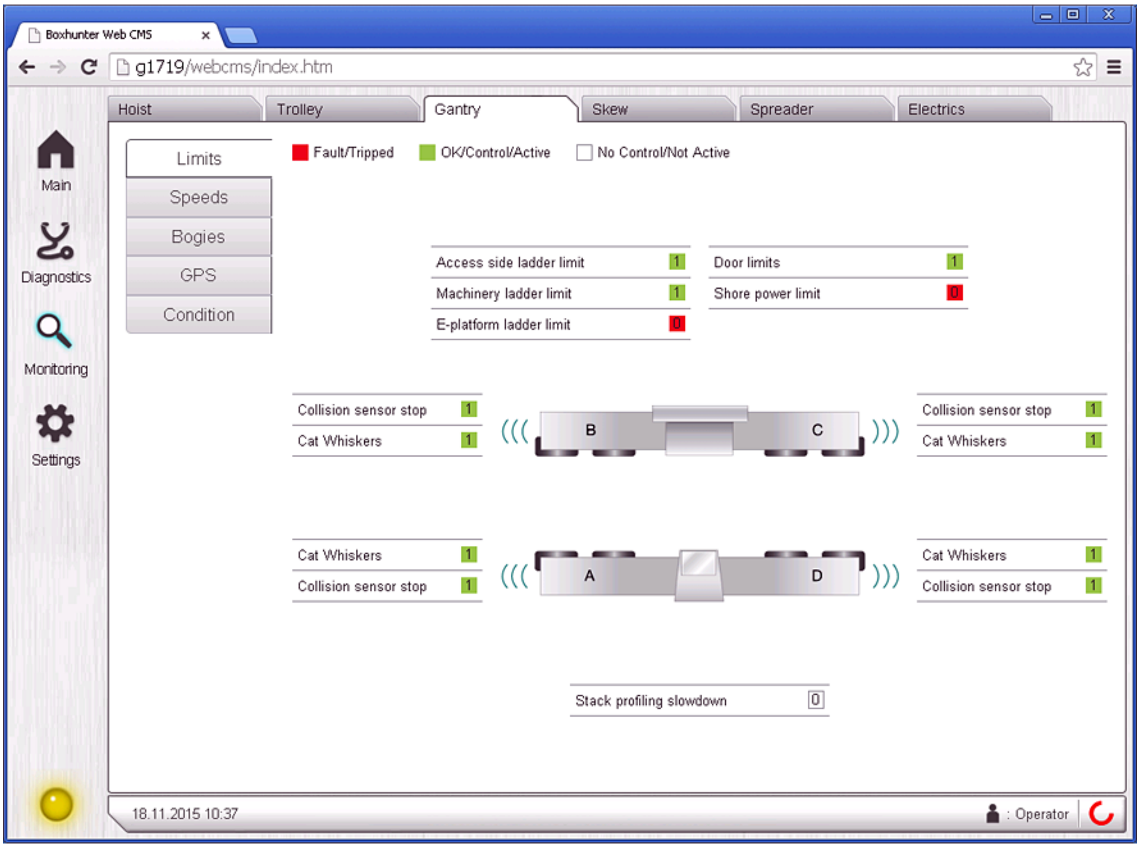


Figure 11. Gantry Limits tab

Gantry Bogies

The *Gantry Bogies* tab displays the bogie locking pin limit states, the bogie angles, and the cross/park/normal states of the bogies.

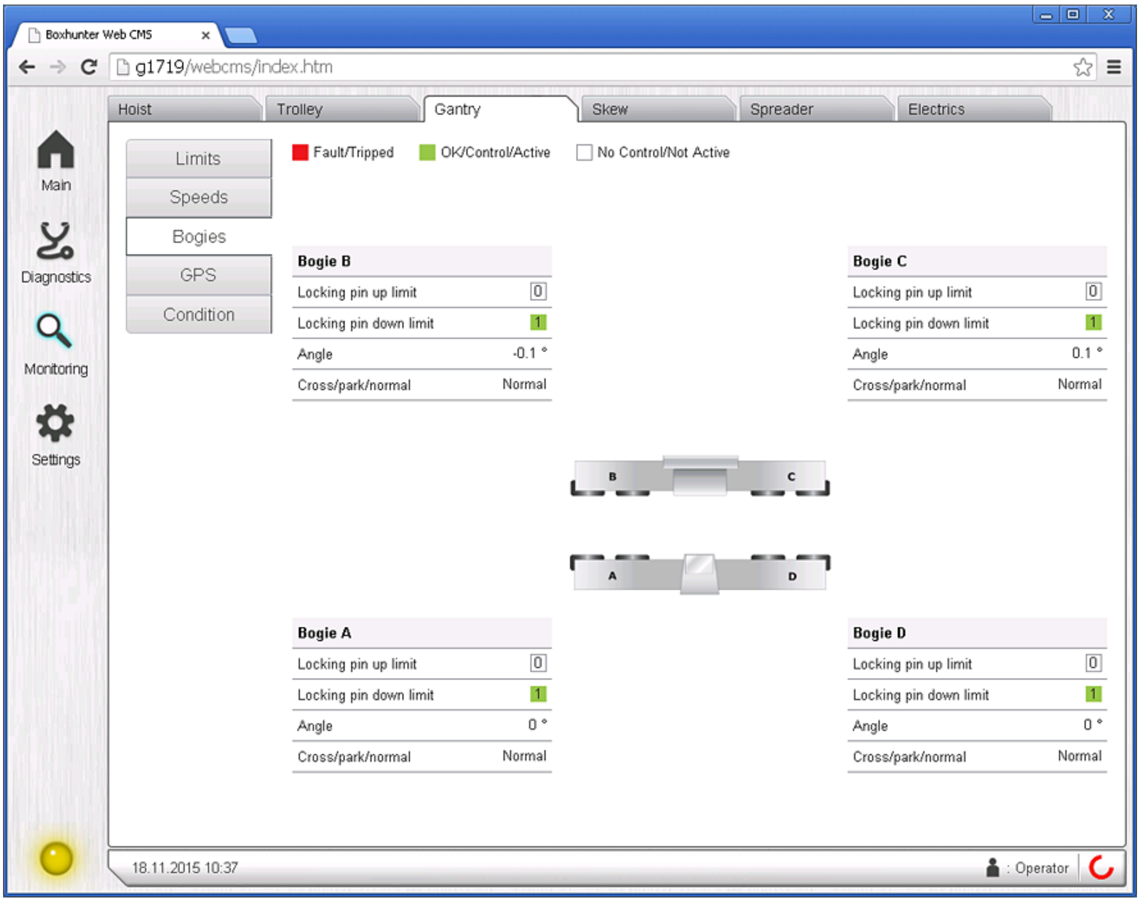


Figure 12. Gantry Bogies tab

Gantry GPS

The *Gantry GPS* tab provides GPS Auto-steering related information. Position data, autosteering states, and the status bits are shown in this tab.

Figure 13. Gantry GPS tab

3.3.5 Skew tab

Skew machinery hardware and software limit states and the skew angle values (calculated, not measured) are shown in the *Skew Limits* tab.

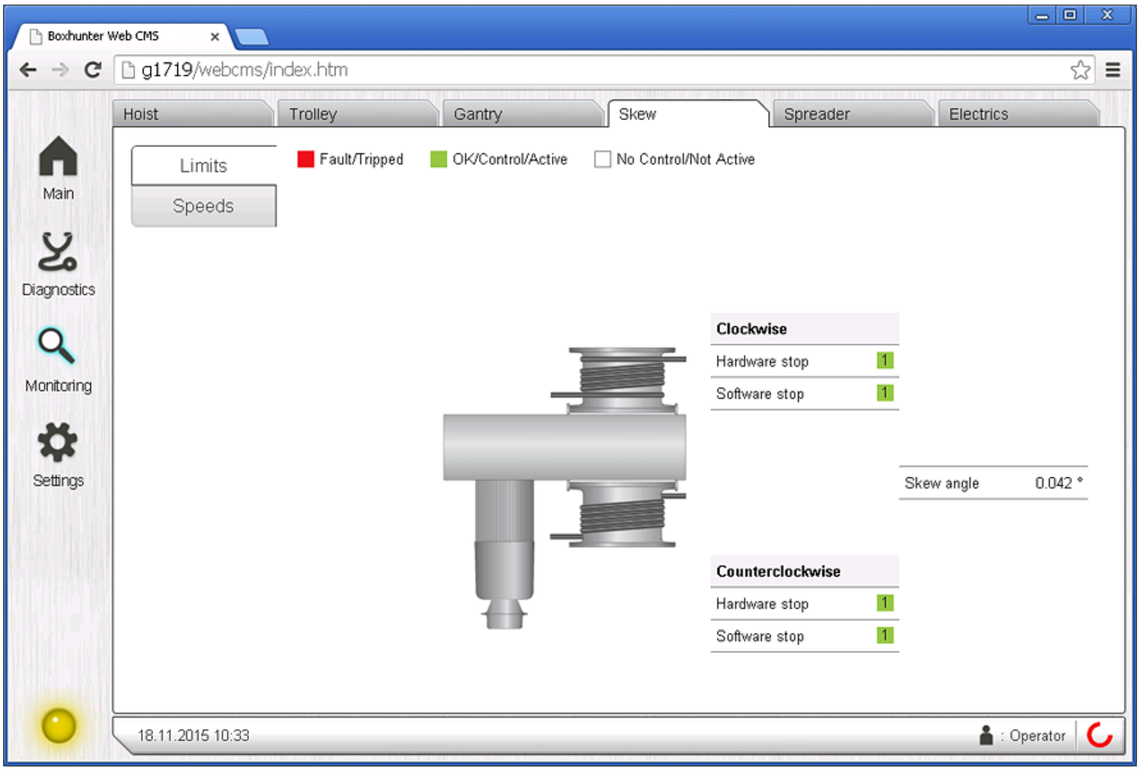


Figure 14. Skew Limits tab

The *Skew Speeds* tab works in the same way as the *Hoist Speeds* tab, but with only one drive and motor.

3.3.6 Spreader tab

The spreader trim and skew angles along with the currently selected spreader telescope length and the twistlock state are displayed in the *Spreader* tab.

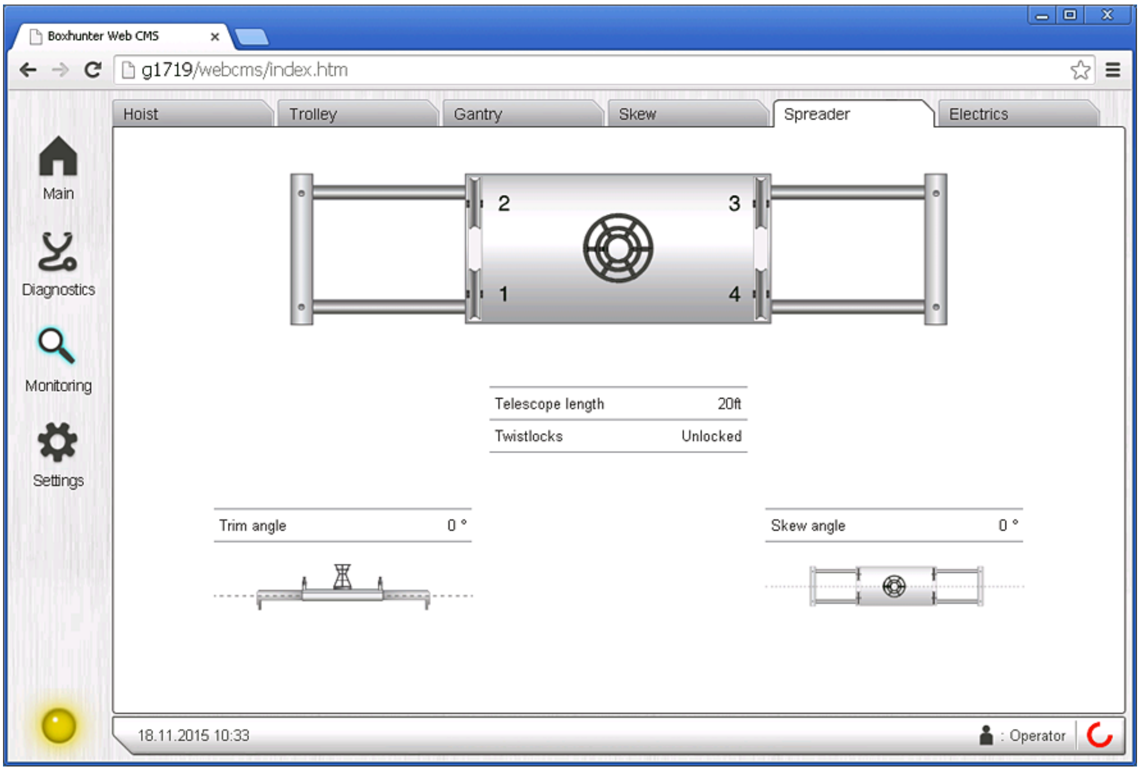


Figure 15. Spreader tab

3.3.7 Electrics tab

The *Electrics* tab of the *Monitoring* page provides information on the crane's electrical systems.

Electrics – E-Stop circuit

The emergency stop circuit is displayed on the *E-stop circuit* tab.

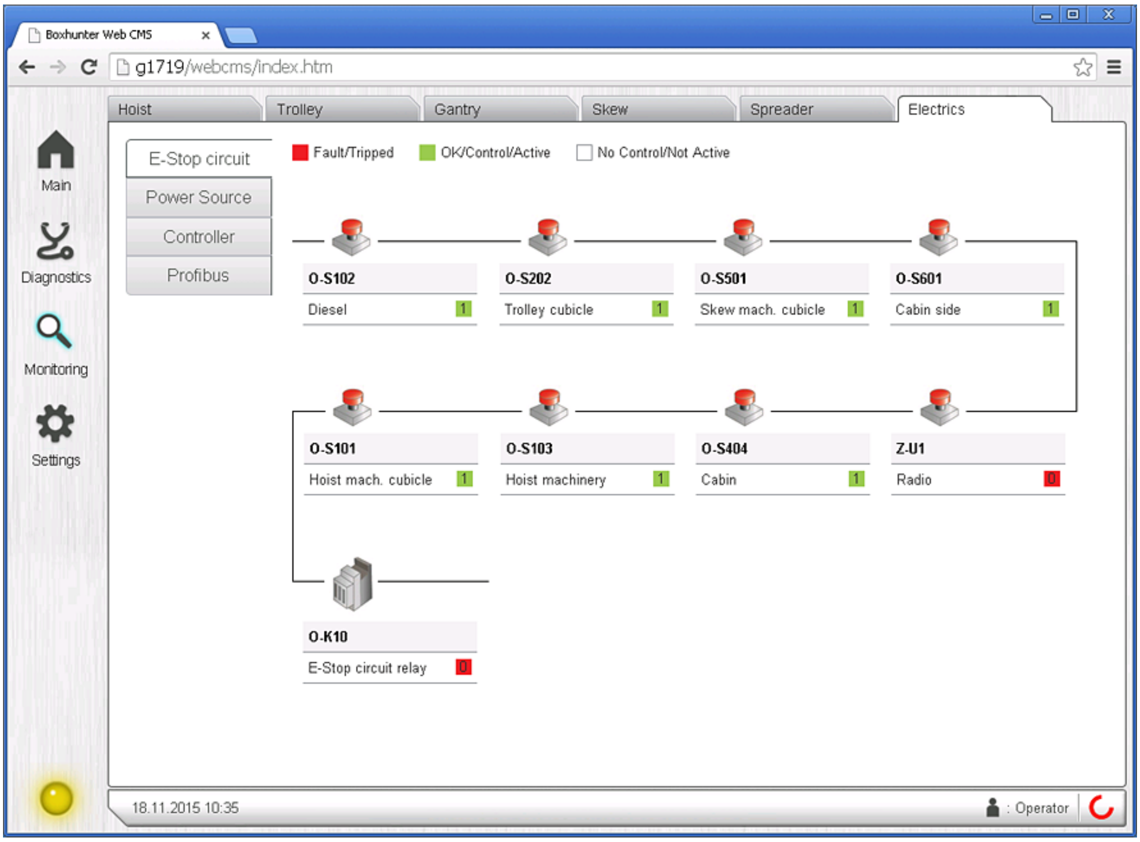


Figure 16. Electrics – E-Stop circuit tab

Electrics – Power Source

The diesel generator states are displayed on the *Power Source* tab.

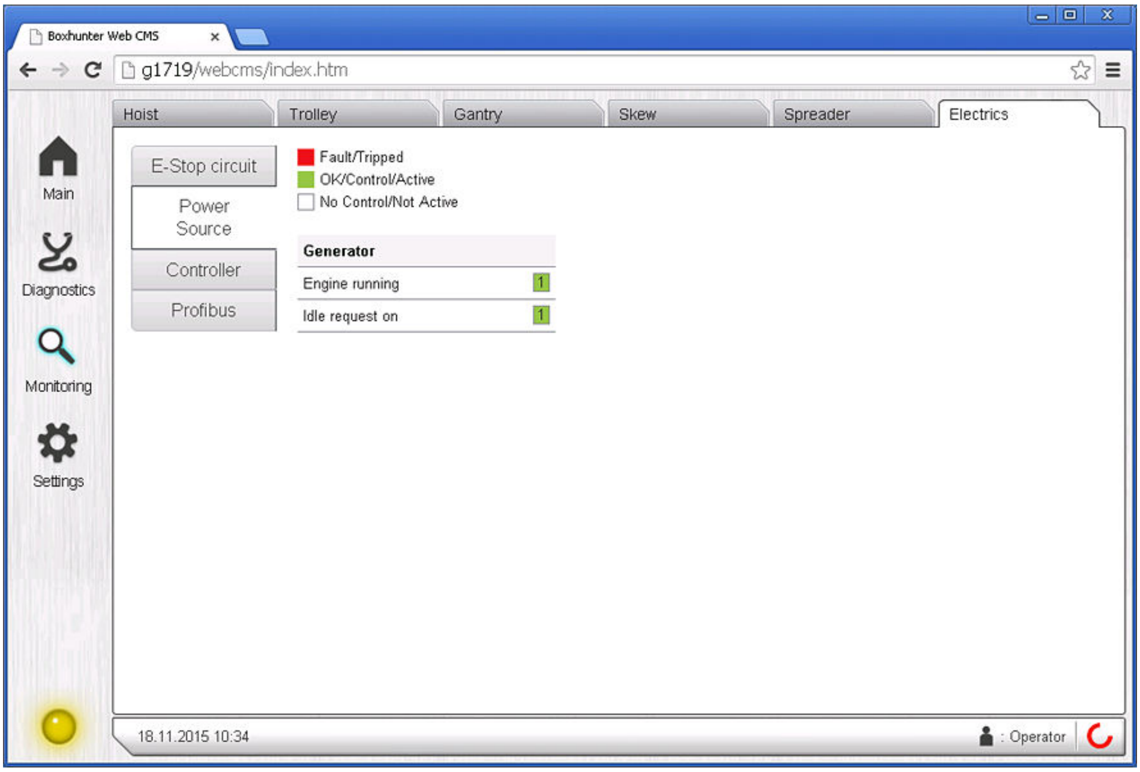


Figure 17. Electrics – Power Source tab

Electrics – Controller

The crane control place can be diagnosed using the *Controller* tab. The tab shows if the control signals are coming in to the control system without problems in cases where the crane has unexpected problems in its operational functions.

The buttons, switches, and joysticks are highlighted in green or change their position on the screen when a control signal is received by the control system. Numerical values are also displayed for speed signals.

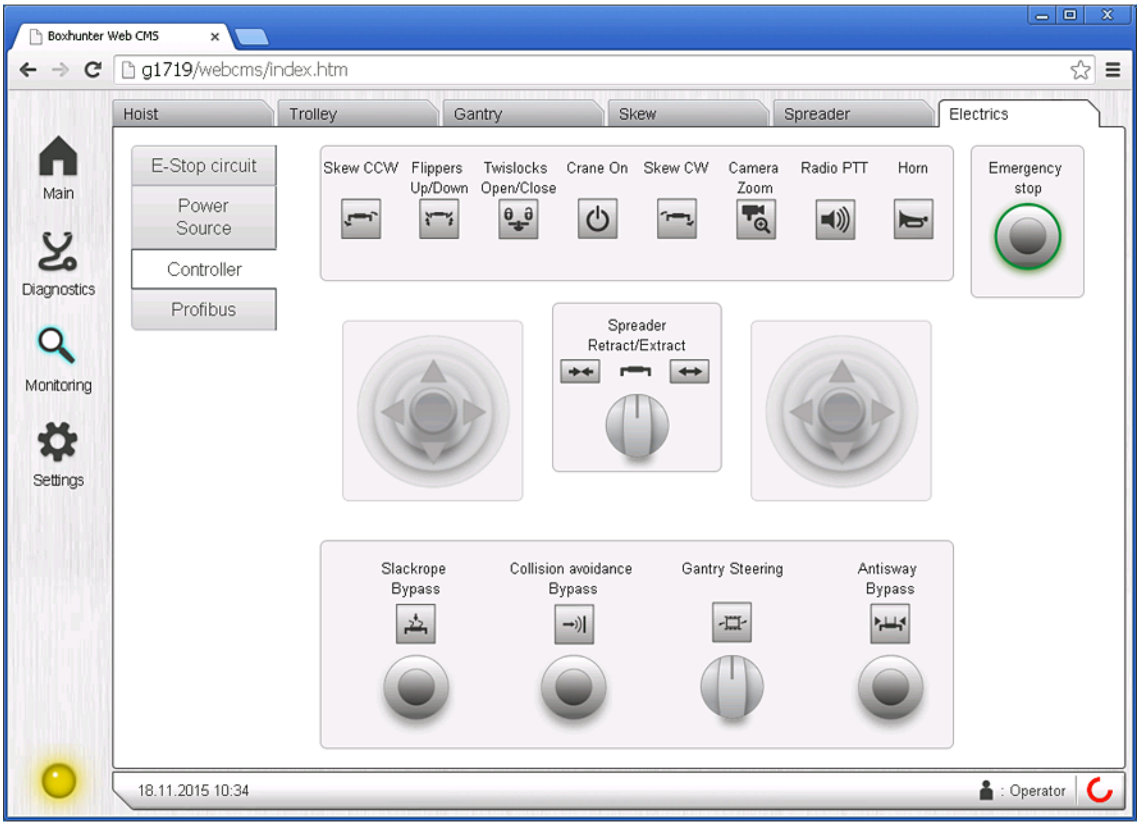


Figure 18. Electrics – Controller tab

Electrics – Profibus

The states of the crane components that are connected via Profibus are displayed on the *Profibus* tab. The information can be used, for example, to diagnose cabling problems. Fiber connections are shown in orange.

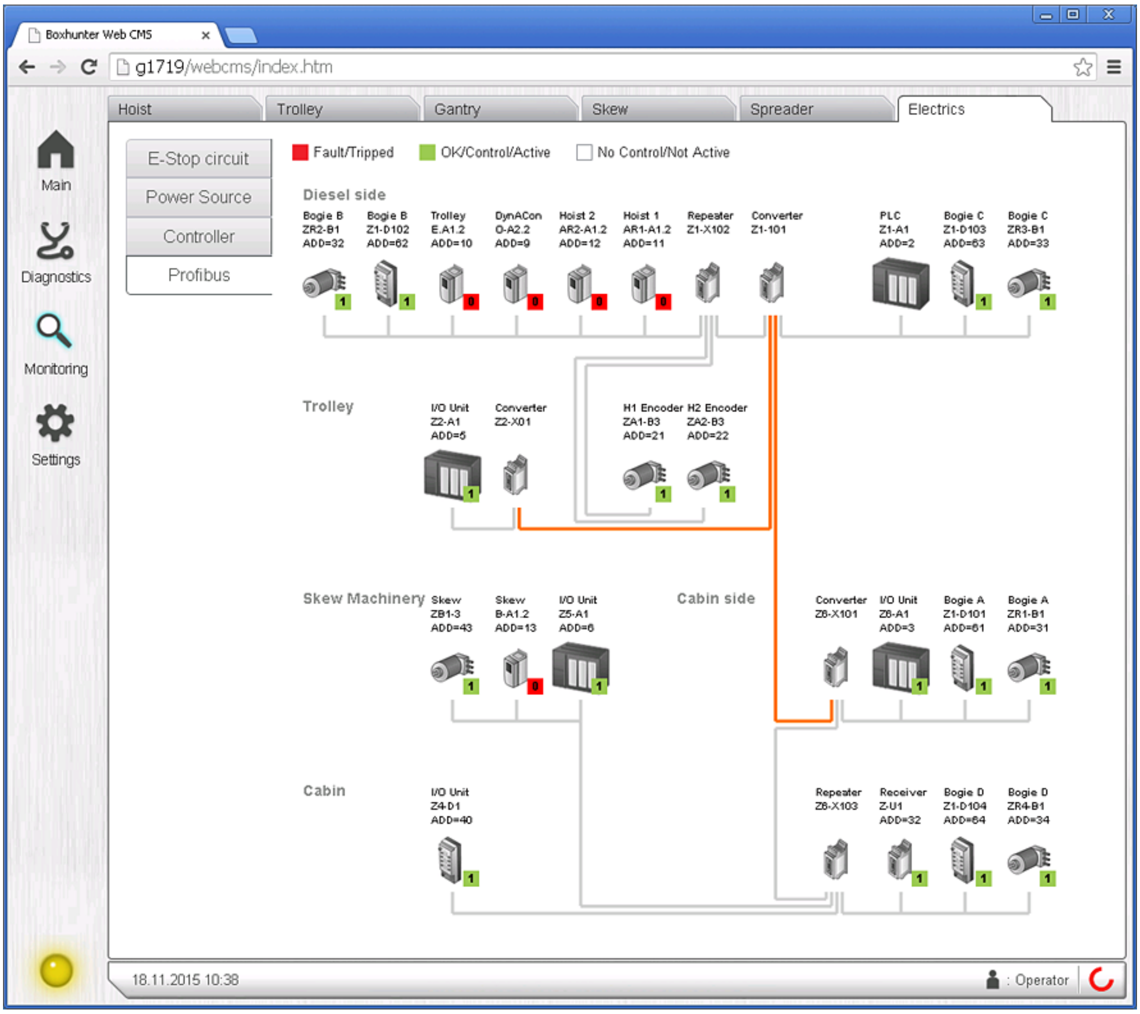


Figure 19. Electrics – Profibus tab

3.4 Settings page

Use the *Settings* page to set the date and time of the control system during commissioning, or whenever such changes are made that reset the date and time to an incorrect value. The page automatically fills in the date and time fields based on information from the computer that is used to connect to Web CMS. To send the values to the control system, press **Apply**.

You can also set the time zone here for remote connection purposes.

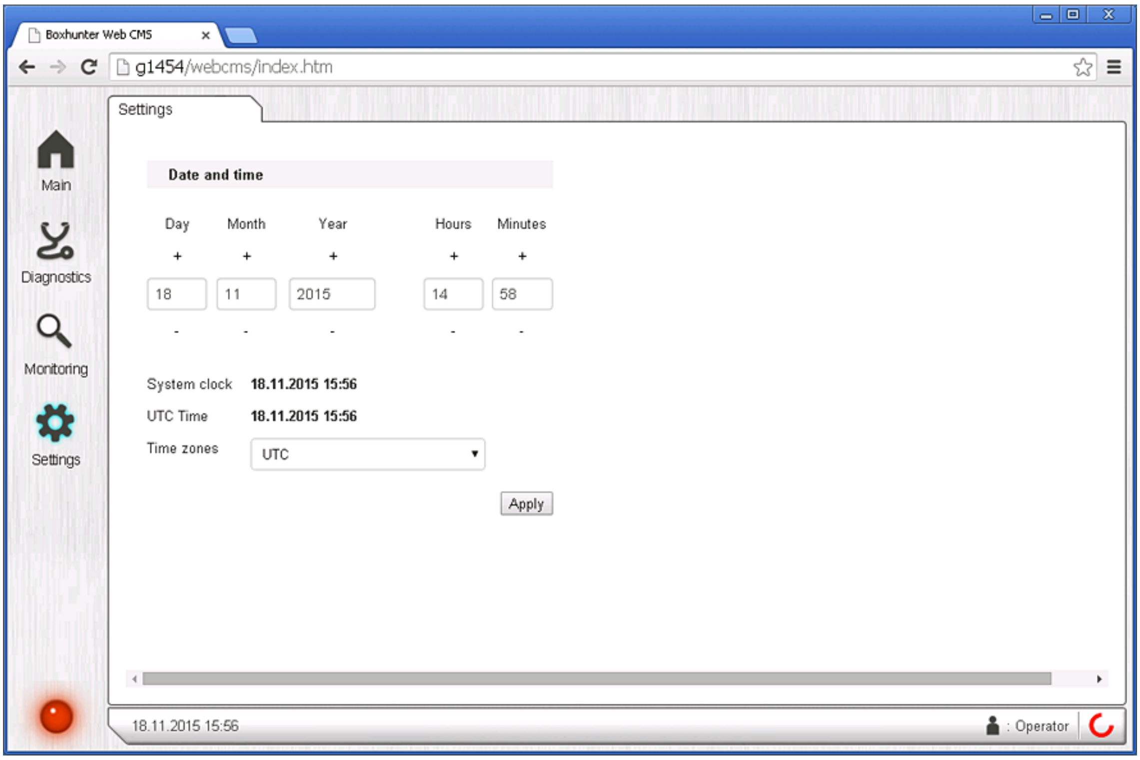


Figure 20. Settings page

4 TROUBLESHOOTING

The connection to Web CMS can be monitored with the control system date and time, which is displayed in the footer of Web CMS. If there are long interruptions with the update of the time values, network problems are the most common cause. In such cases, check all the cables connecting your computer to the crane and the Web CMS network. Also check the signal levels if you are using a wireless connection.

If there is a connection failure:

- Check that the crane is online and that the connection between the crane and the client machine is working.
- If there are many concurrent connections to Web CMS, close all the other connections.
- If the crane has not been accessible from CMS Server for a long time, wait for Data Logger to transfer all its data to CMS Server.

Web CMS displays an error pop-up window when it detects an internal problem, such as communication errors with the crane control system. In such cases, store the error pop-up message and contact your local Konecranes Service or the crane commissioning personnel.

If any other errors occur, check that:

- Your web browser has JavaScript enabled
- Web CMS supports you web browser version (two latest versions of Internet Explorer, Google Chrome and Firefox)

If there is an unexpected error, clearing your web browser's cache and restarting it may help.

5 DESCRIPTION OF THE WEB CMS COMPONENTS

This chapter gives a short introduction to the various electrical components used in the Web CMS system.

5.1 C6915 Industrial PC

The Web CMS system's CMS Server, Web Service, and Data Logger functions run on a Beckhoff C6915 Industrial PC (IPC). Its integrated Ethernet connection is used to communicate with the Siemens S7 PLC of the crane and other devices in the crane's internal network.

To deploy software to the device, flash an image to the CF (Compact Flash) memory card, insert the card to the IPC, and power the device on.

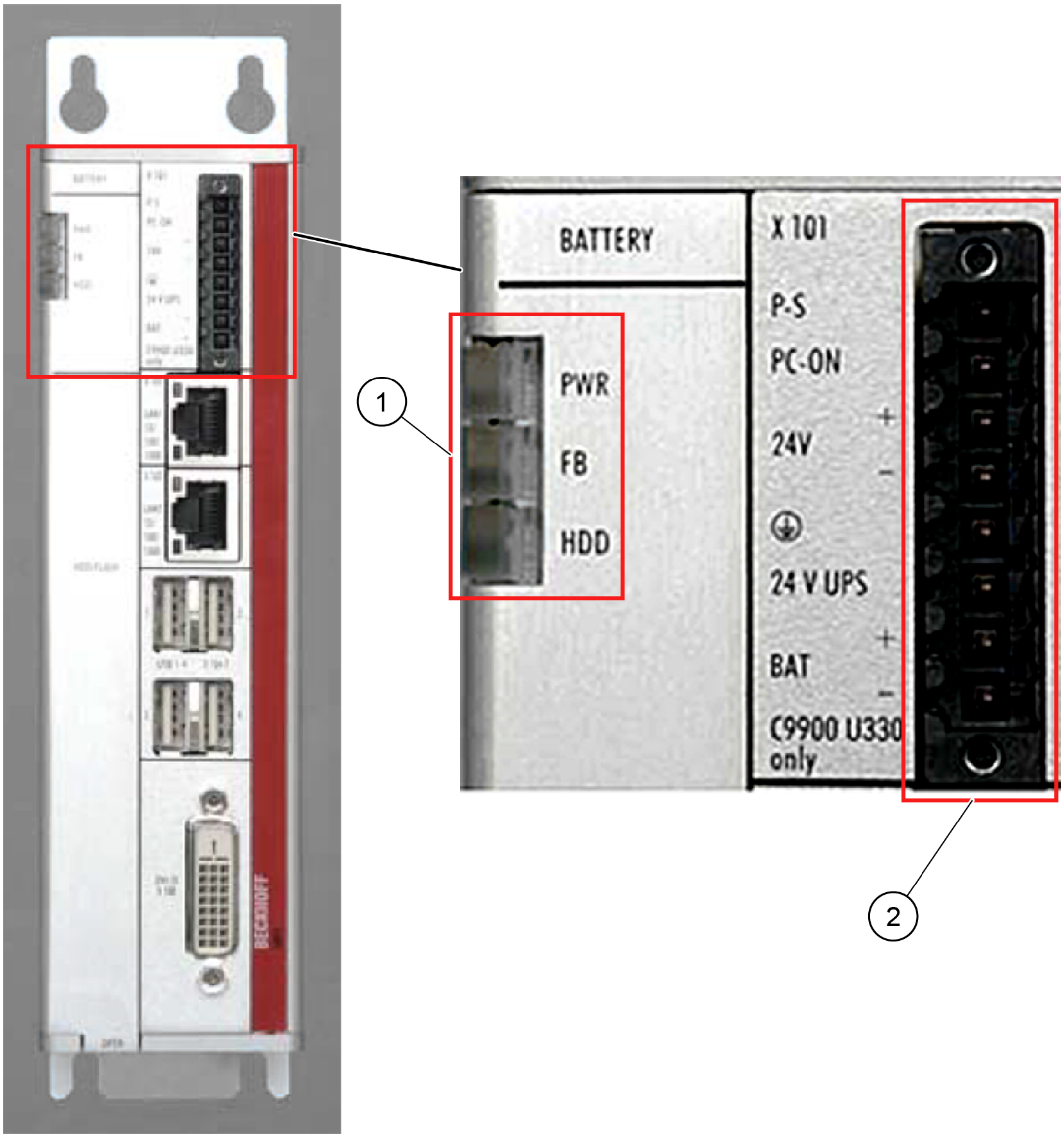


Figure 21. Beckhoff C6915 Industrial PC status LEDs and power supply connector

1. *Status LEDs*
2. *Power supply connector (X101)*

The following table illustrates the Beckhoff C6915 Industrial PC status LEDs:

PWR (Power):	Green	Industrial PC is busy
	Off	Industrial PC is off
FB (Fieldbus):	Red	TwinCAT stop mode
	Blue	TwinCAT configuration mode
	Blue / red blinking	TwinCAT Config (fieldbus error)
	Green	TwinCAT run mode
	Green / red blinking	TwinCAT Run (fieldbus error)
HDD (Hard disk):	Red	Access to memory device

5.2 SITOP UPS500 DC UPS

A SITOP UPS500S DC UPS provides uninterrupted power to the Beckhoff IPC and ensures its controlled shutdown if there is a power outage. The UPS500S is a 24 V DC capacitor UPS module which provides power and a hardwired shutdown signal (via a time relay) to the IPC.

Technical data[1]		
Basic units	**SITOP UPS500S - 15 A**	
	2.5 kWs	**5 kWs**
Order number[2]	6EP1 933-2EC41	6EP1 933-2EC51
Input data		
Rated input voltage $U_{in\ rated/\ Range}$:	**24 V DC** / 22 ... 29 V	
Connection threshold for buffering	22.5 V DC ± 0.1 V (factory setting), 22 ... 25.5 V DC in 0.5 V increments, adjustable	
Input current $I_{in\ rated}$	15.2 A + approx. 2.3 A in charging mode	
Mains buffering		
Adjustable range using DIP switches	5, 15, 25, 35, 45, 55 etc. up to 315 seconds (in 10-s increments) or max. buffering time	
Behavior on restoration of input voltage after buffering time	Interruption of U_{out} for 5 s for the automatic rest	
On/off control circuit (via external floating NO contact)	by opening the circuit the buffer mode is terminated	
Output data		
Output voltage in normal operation	24 V DC +/-3%	
Output voltage in buffering mode	24 V DC +/-3%	
Output current I_{out}	**0 ... 15 A**	
Charging current	1 A (factory setting) or 2 A, selectable	
Efficiency / power loss approx.	97.5% / 9 W (rated operation)	
Protection and monitoring		
Polarity reversal protection	against polarity reversal on input voltage	
Overload / short-circuit protection	yes, restart	
Normal operation	LED green (OK) and floating changeover contact	
Buffering mode	LED yellow (Bat) and floating changeover contact	
Alarm (not ready for buffering)	LED red (Alarm) and floating changeover switch	
Charging status (over 85% charged)	Second LED green and floating NO contact closed (Bat > 85%)	

Figure 22. SITOP UPS500S DC UPS technical data

5.3 Connecting the Beckhoff Industrial PC

Make sure that the Beckhoff C6915 is assembled and connected as shown in the following figure. It is located in the BOXHUNTER diesel-side cubicles.

If the system is powered on, the status indicators may show an error. This is normal. The condition should be automatically rectified when the Web CMS software is flashed to the CF memory card and the system is restarted.

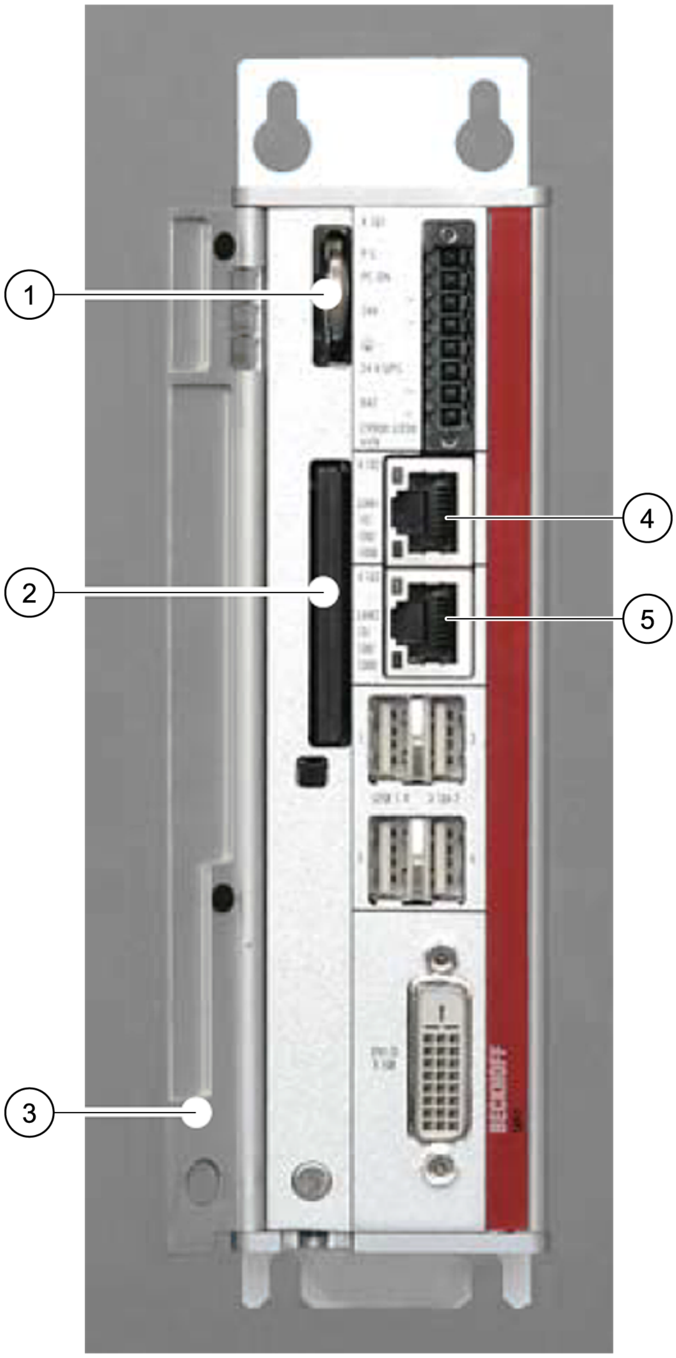


Figure 23. Beckhoff C6915 connections and components

1. *CMOS battery*
2. *CF memory card*
3. *Front flap*
4. *Connected to PLC via Ethernet switch*
5. *Connected to customer wireless network (optional)*

6 FLASHING THE CF CARD WITH THE WEB CMS SOFTWARE

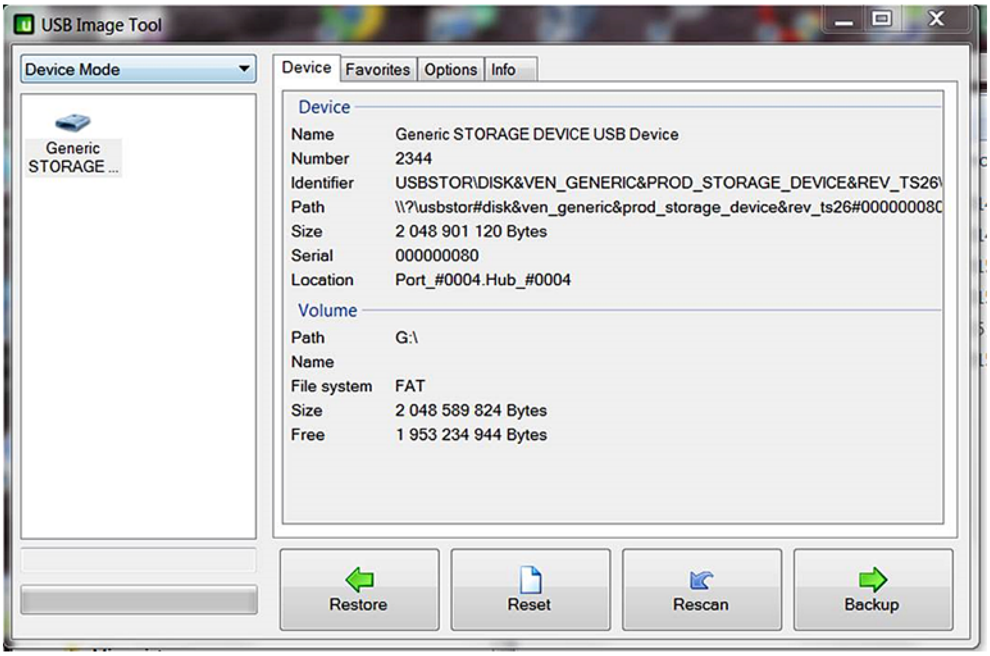
Needed hardware:

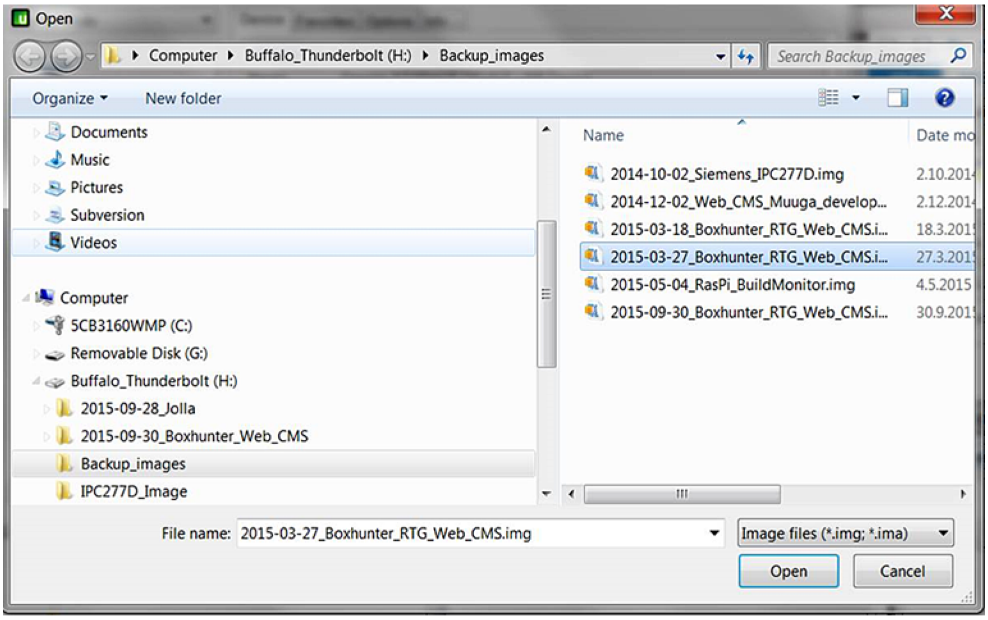
- CF memory card reader
- PC compatible with the card reader
- Flat-headed screwdriver for CF card removal

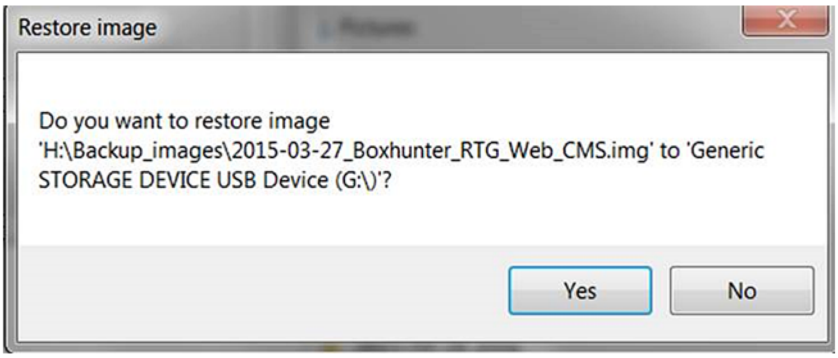
Figure 24. CF memory card and reader

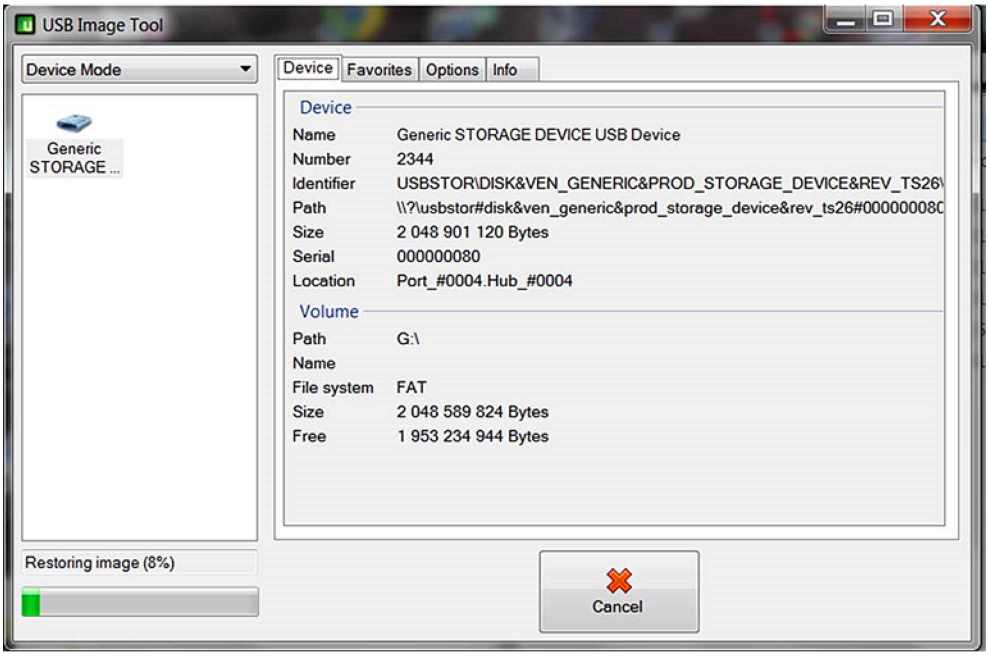
1. *Disconnect power from the Beckhoff C6915 system.*
2. *Remove the CF card from Beckhoff C6915 and insert it in the card reader.*

3. *Flash the Web CMS software to the CF card. This can be done using, for example, USB Image Tool.*









4. *Insert the card in Beckhoff C6915 and power on the device. The Web CMS software should start automatically.*

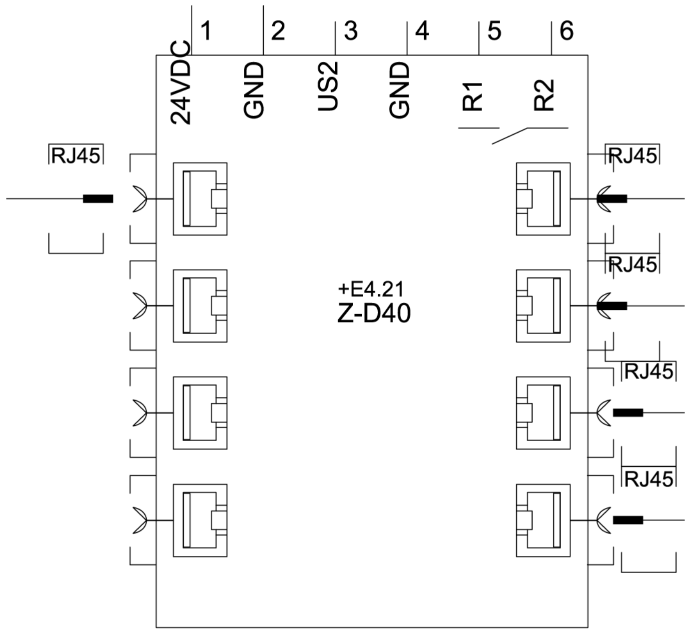
NOTE ***First time start-up takes longer than usual, typically 2 - 4 min.***

7 TESTING WEB SMC WITH A WEB BROWSER

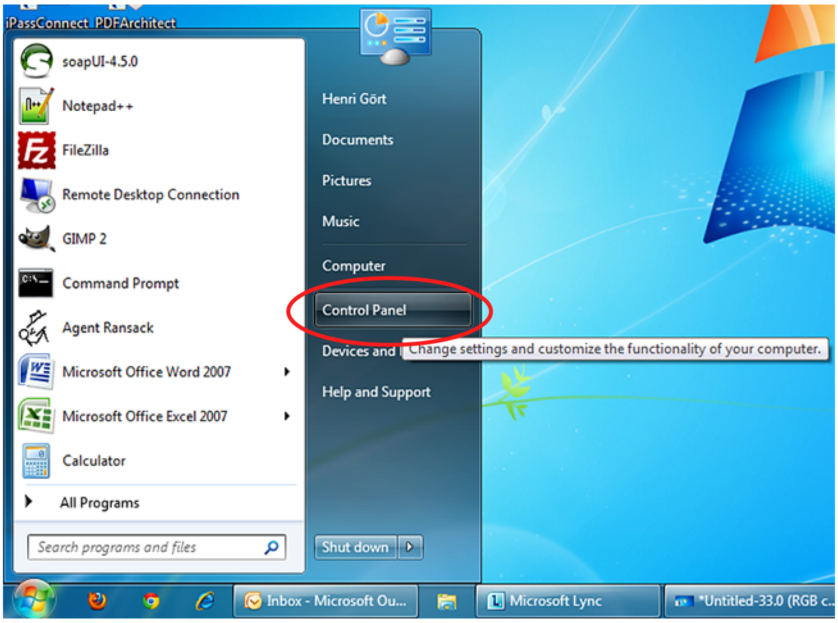
This chapter describes how to test the Web CMS system once you have completed all the steps mentioned in the overview.

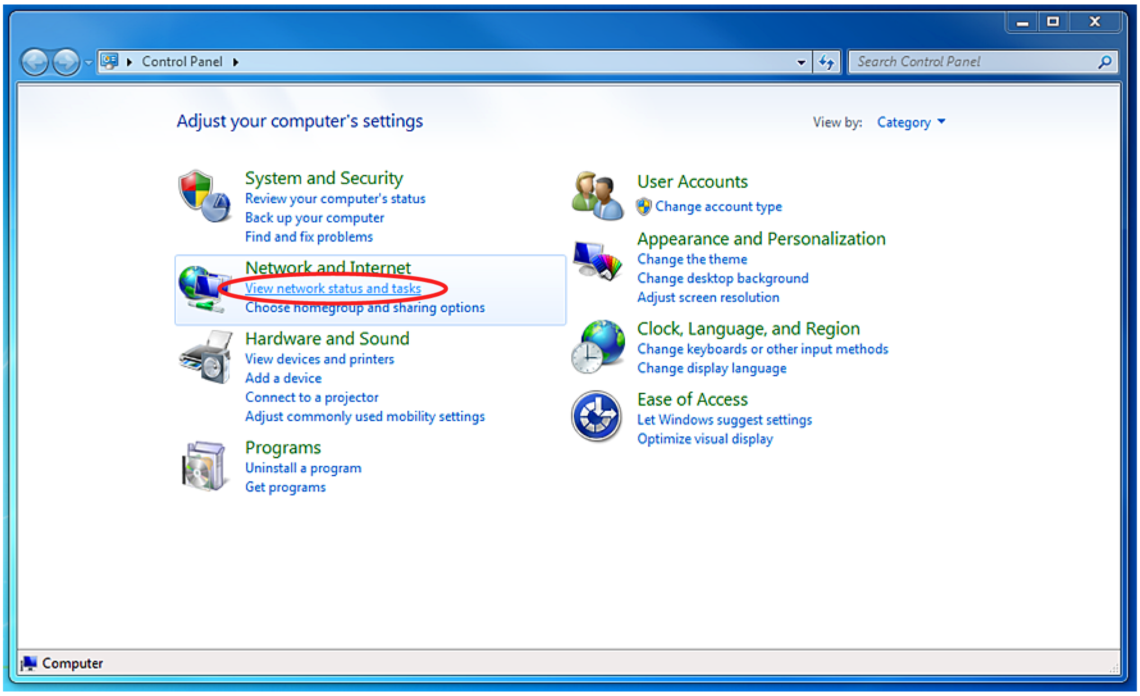
7.1 Connecting the PC to the crane network

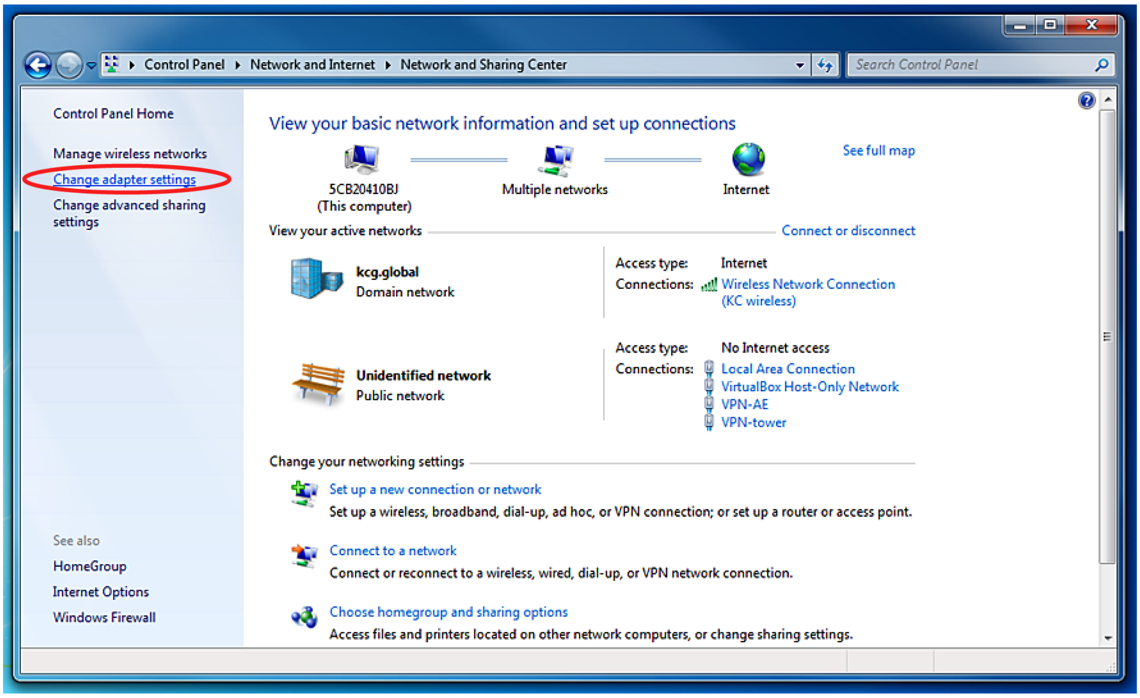
1. *Connect the Ethernet cable to the PC's Ethernet port and an available port on the crane's Ethernet switch.*



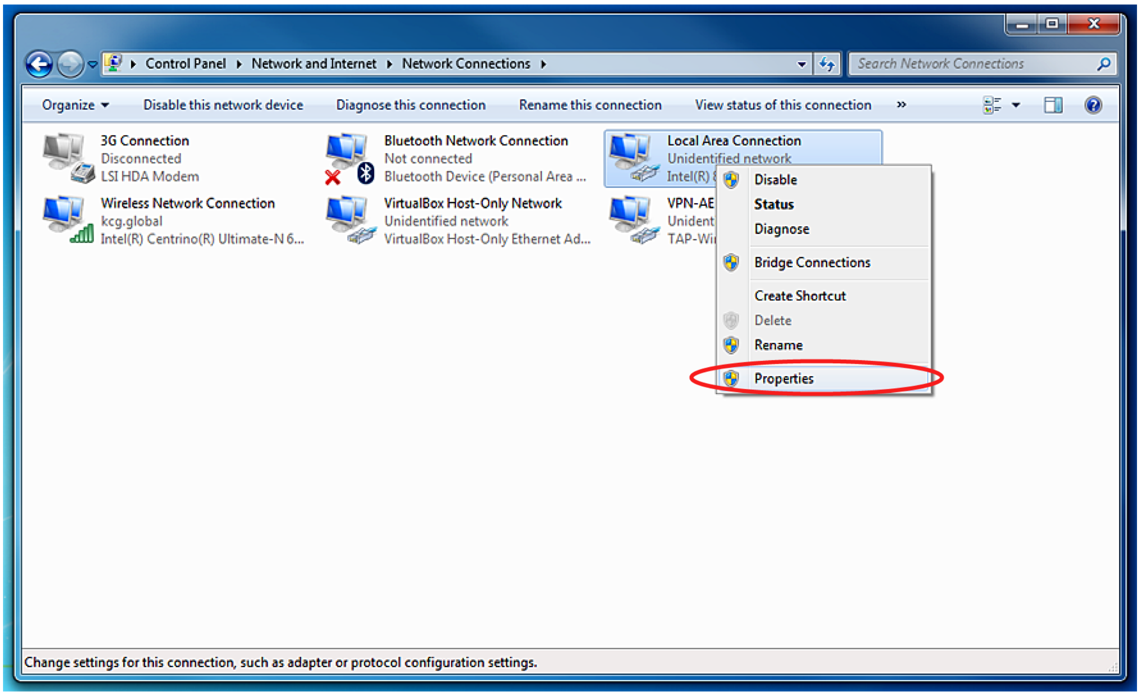
2. *Open the PC's network connections.*



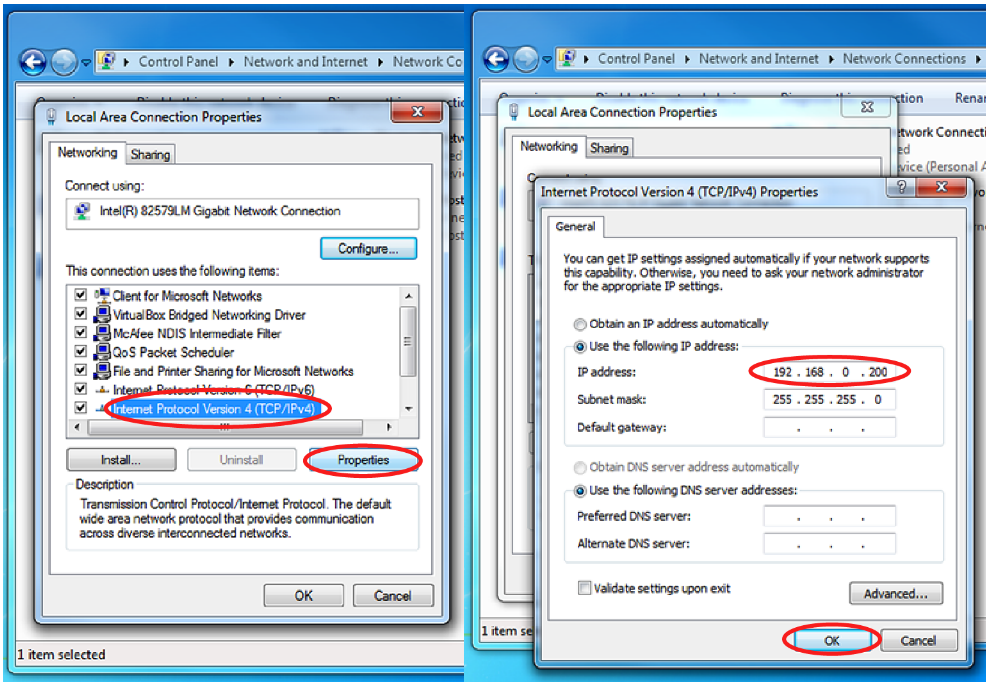




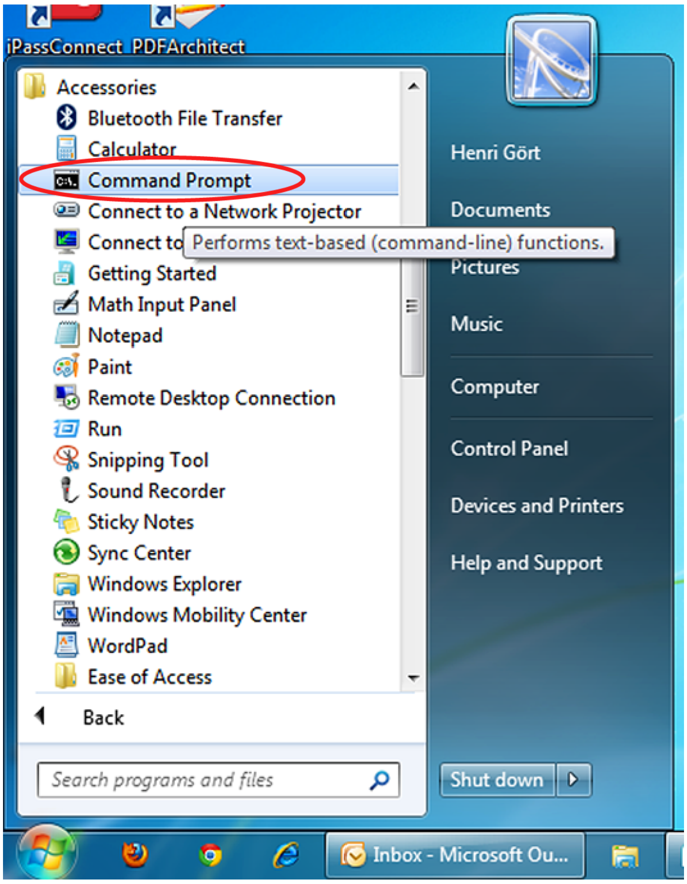
3. *Right-click on **Local Area Connection** and open **Properties**.*



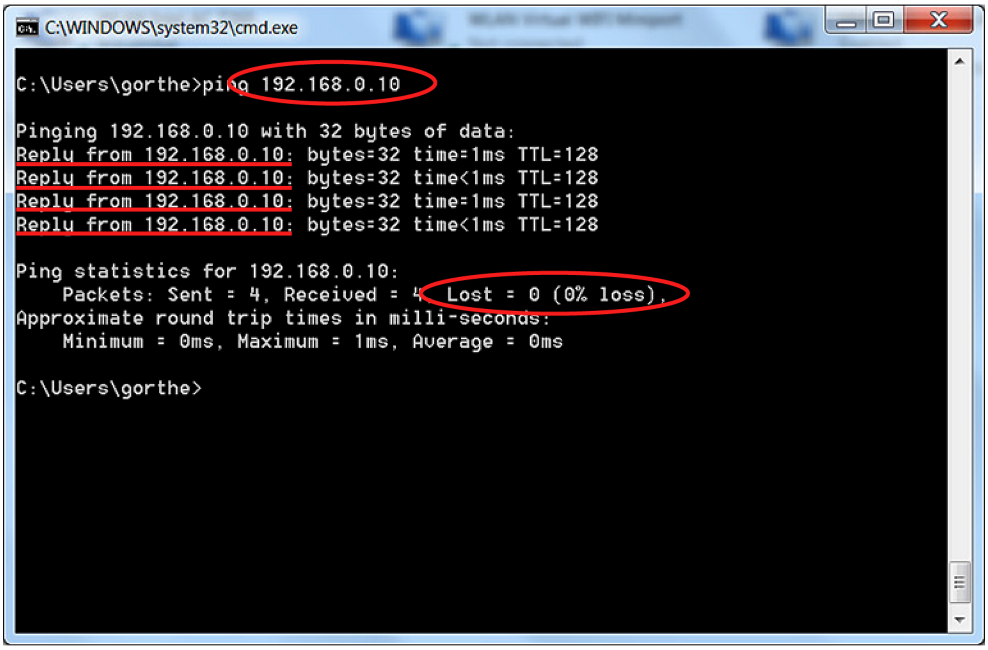
4. *Open the Internet Protocol (IPv4) properties and enter `192.168.0.200` as the IP address. Subnet mask is set automatically.*



5. *To apply the settings, press* ***OK****.*
6. *Open* ***Command Prompt****.*

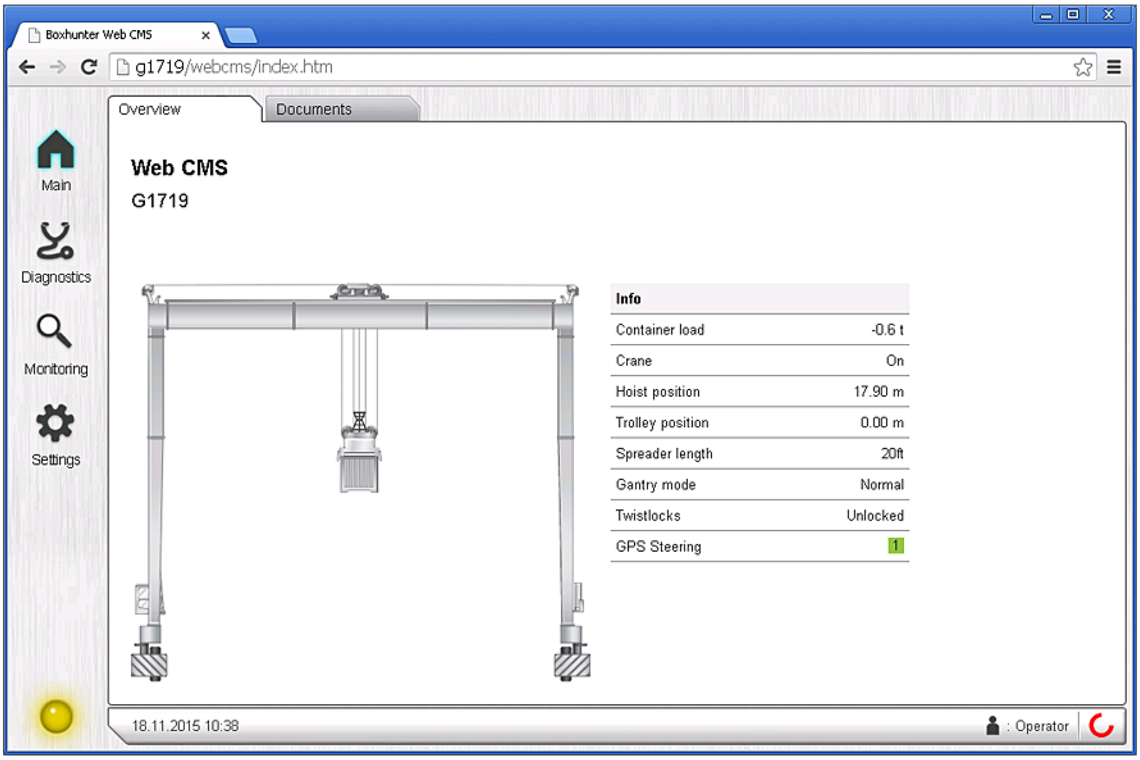


7. *To verify that you have a connection to the Beckhoff IPC, enter the command ping* `192.168.0.10` *and make sure that you get replies and no errors.*



7.2 Opening Web CMS on the PC

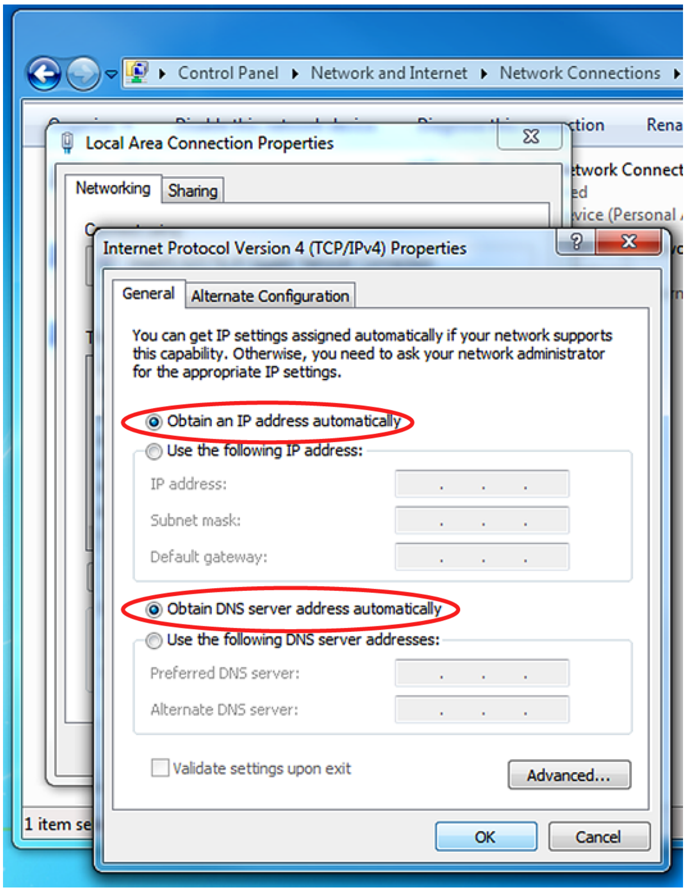
1. *Start the PC's web browser and enter* `http://192.168.0.10/webcms/` *in the address bar. A page similar to the following should be displayed:*



2. *Test that the Web CMS works correctly:*
   - *The Web CMS page loads in the web browser and displays values.*
   - *There are no PLC communication errors if the PLC is connected and powered on.*

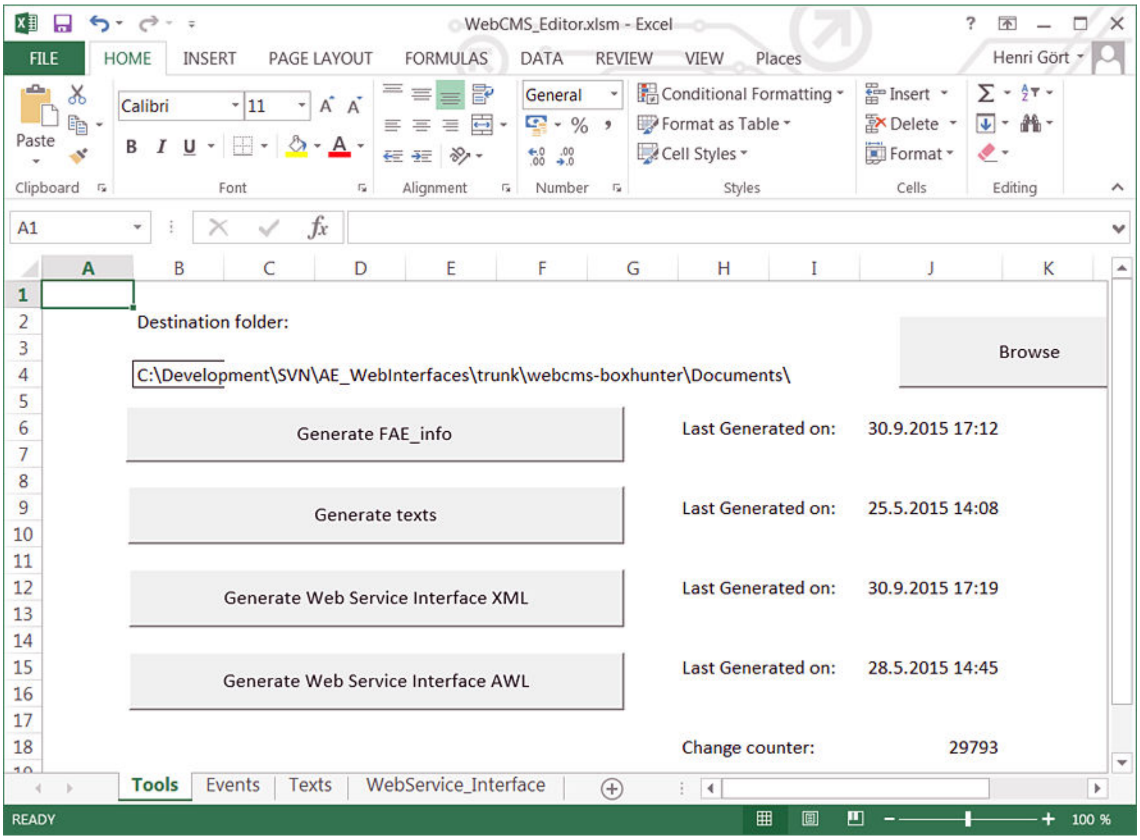
   Test the Web CMS system with at least one of the following web browsers:

   - *Mozilla Firefox version 5 or newer*
   - *Google Chrome version 13 or newer*
   - *Microsoft Internet Explorer 8 or newer*
3. *If the test was successful, return the PC's network settings to their previous state by following the instructions in* *Connecting the PC to the crane network (page 31)* *and setting the IPv4 properties to obtain IP address and DNS servers automatically (DHCP):*

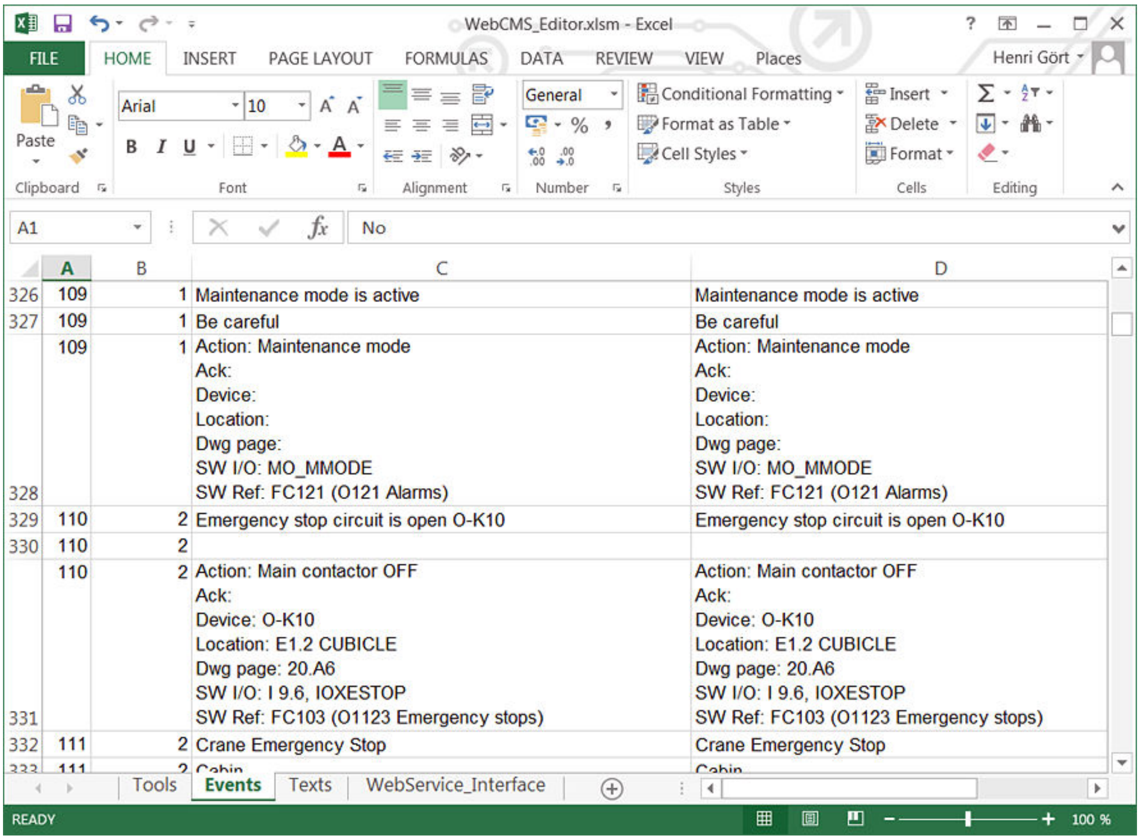


8 UPDATING THE MESSAGE DESCRIPTIONS

The message description texts and the message classes (Fault, Alarm, Event and Bypass) are entered in a tool called Web CMS Editor. The editor is a macro-enabled Excel spreadsheet containing, for example, message information on separate sheets. The sheet row and column formatting is identical to the traditional CMS Editor tool used by Konecranes, which enables simple transfer of text and other message information between these two tools (for example, by copying and pasting the entire sheet from one tool to the other).



Once you have entered the message information, deploy it in Web CMS by generating a file named `FAE_info.xml` from the editor and transferring it to Web CMS IPC. To generate the file from the Tools sheet, configure the correct destination folder and press **Generate FAE_info**.



You can then transfer the file to the Web CMS memory card using, for example, the memory card reader used in Flashing the CF card with the Web CMS software (page 28). Transfer the file to the `inetpub\wwwroot\` directory on the card.

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